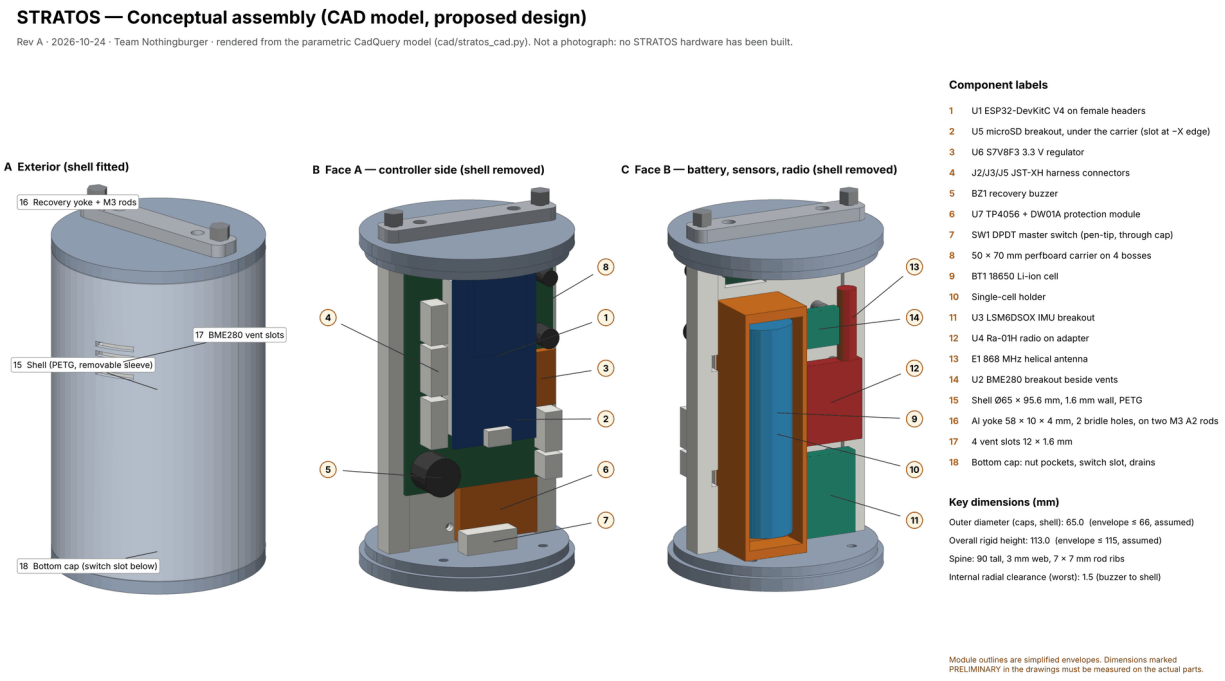
STRATOS

Engineering Dossier — environmental sensing, onboard logging, telemetry and recovery payload

SEVA FIRST Innovation Challenge 2026 · Team Nothingburger



Team: Nothingburger (member names to be added by the team)

Competition: SEVA FIRST Innovation Challenge 2026 — *Diploma & Higher Education* track;

Document: STRATOS-ED

Project type: CanSat-inspired environmental sensing, telemetry, onboard data logging and recovery system

> **STATUS: PROPOSED ENGINEERING DESIGN. No STRATOS hardware has been purchased, assembled, integrated or flown. The software pipeline has been exercised with synthetic (simulated) data only. The rendered assembly above is a CAD model, not a photograph.**

Executive summary

STRATOS is a proposed can-sized payload that measures air temperature, humidity, pressure and motion, keeps a complete copy of every measurement on board, sends a structured subset to a ground station by radio, and — after recovery — proves which data arrived, which did not and why. Its distinguishing idea is **data integrity as a design objective**: the payload record and the ground record are linked by session IDs, record and packet sequence numbers, CRC checks and an explicit reconciliation procedure, so that losses are measured rather than hidden behind a live graph.

Table 0-3 Baseline design at a glance (Rev A)

Item	Baseline (PROPOSED)
Controller	Espressif ESP32-DevKitC V4 (ESP32-WROOM-32E), Wi-Fi/Bluetooth off, CPU at 80 MHz
Sensors	Bosch BME280 (T, RH, P) and ST LSM6DSOX 6-DoF IMU on I2C; battery voltage on ADC1
Onboard storage	microSD (3 V SPI breakout), one CSV line per 10 Hz record, each line protected by its own CRC-16
Telemetry	Ai-Thinker Ra-01H (Semtech SX1276), GFSK 866.0 MHz, 38.4 kb/s, +14 dBm, 1 packet/s, 44-byte packet with CRC-16, 11.5 ms on air (1.15 % duty)
Ground station	Second Ra-01H + ESP32 receiver over USB to a laptop running the stratos_gs Python software
Power	One 18650 Li-ion cell, DW01A protection, PTC fuse, DPDT master switch, Pololu S7V8F3 3.3 V buck-boost; estimated 52 mA average → ≈43 h with the BOM cell (2,800 mAh listing value, 80 % usable) against a 4 h requirement
Structure	PETG caps, spine and sleeve; two M3 stainless rods carry recovery loads; Ø65 × 113 mm, inside the assumed Ø66 × 115 mm envelope
Recovery	Ø0.33 m flat circular parachute on a two-leg bridle; ≈9 m/s at the 325 g design mass (calculated, C_D assumed 0.75)
Mass / cost	≈243 g before ballast (estimate); payload production ≈₹11,053, total procurement incl. ground station, spares and 15 % contingency ≈₹19,150 (prices read 24 Oct 2026)

What exists today. A complete design package generated from one set of design values, so the documents agree with each other: requirements, architecture, CAD model and dimensioned drawings, a netlist-generated schematic and pin map, power/storage/mass/link/parachute calculations, an itemised BOM with sourced prices, a risk register, a roadmap and a verification matrix. An early benchwork prototype. A working **software prototype** of the packet codec, ground-station validation and post-mission reconciliation has been built and tested (8 unit tests pass; a C implementation of the packet code matches the Python byte-for-byte). It has only ever processed **synthetic data**. Some sensors and electronic components have assembled.

What does not exist yet. No sensor, storage, radio, power, mechanical, recovery or flight test has been performed. The payload firmware is an uncompiled draft.

Current project maturity statement

Progress is stated as completed, evidenced items — not as a percentage.

Table 0-4 Maturity by work area (24 Oct 2026)

Work area	Status	Evidence
Concept, mission and requirements	Draft complete — team review pending	Chapter 1; TVM workbook
System architecture and interfaces	Draft complete	System architecture SVG; netlist; pin map
Mechanical concept and drawings	Draft complete (PRELIMINARY module outlines)	CAD model, STEP files, 4 drawing sheets, CAD check report
Electronics schematic	Draft complete; not reviewed by an electrical engineer; no PCB	3-sheet schematic, KiCad-format netlist
Firmware	Architecture and state machine defined; packet/log code host-tested; main sketch uncompiled	State diagram; firmware/ folder
Ground-station and reconciliation software	Prototype complete — synthetic data only	8 unit tests; simulated reconciliation report
Calculations and budgets	Preliminary — many inputs ASSUMED	Power, storage, mass, link, parachute workbooks
BOM and cost	Prices sourced 24 Oct 2026; nothing ordered	BOM workbook (all lines 'Proposed')
Component procurement	Early stage	—
Physical assembly and integration	Early stage	—
Sensor, storage, telemetry, integration tests	Partky assembled and some still acquiring	—
Mechanical, recovery and flight tests	Not started; no permissions held	—
Reference footage	External material only — not STRATOS evidence	Footage register

1 Mission and requirements

1.1 Problem definition

A small sensor payload can show convincing live graphs and still fail its purpose: readings may lack timestamps, packets may be lost without trace, the ground record may be incomplete, and the recovered payload may hold data that nobody compares with what was received. STRATOS treats measurement as a chain — measurement → acquisition → local recording → transmission → reception → verification → reconciliation — and gives every link a defined responsibility and a test.

The design keeps five claims separate, because each needs different evidence:

Table 1-1 The five claims kept separate

Claim	Question it answers	How STRATOS addresses it
Measurement validity	Is the reading physically credible?	Plausibility limits, status flags, calibration comparison (FR-01, FR-02)
Record validity	Is the stored record structurally intact?	Per-line CRC-16 in the onboard CSV (FR-05)
Transmission integrity	Was the packet corrupted?	CRC-16 checked before any field is used (FR-07)
Reception completeness	Did every expected packet arrive?	Packet sequence numbers, gap log (FR-08)
Dataset reconciliation	Do onboard and ground records agree?	Field-by-field comparison and classification (FR-09)

A CRC only detects corruption; it says nothing about sensor accuracy. Receiving every packet does not prove calibration. These limits are stated wherever the integrity features are described.

1.2 Mission objectives

Table 1-2 Mission objectives and baseline response

#	Objective (from the brief)	Baseline response
1	Environmental sensing	BME280 temperature, humidity, pressure at 10 Hz. Particulate sensing excluded (§4.1); GNSS optional (pins reserved)
2	Embedded data acquisition	Fixed 100 ms cycle; session ID, rec_seq, elapsed time, status flags in every record
3	Independent onboard logging	microSD CSV, written whether or not the radio works, flushed every second
4	Wireless telemetry	SX1276 GFSK at 866 MHz, 1 packet/s, within the India 865–867 MHz licence-exempt limits
5	Data-integrity verification	CRC-16, sequence tracking, reject log, post-mission reconciliation (software prototype exists)
6	Mechanical protection and recovery	Rod-and-spine structure, removable sleeve, Ø0.33 m parachute on a two-leg bridle
7	Ground-station operation	Receiver + Python software: validation, logging, gap display, export, reconciliation

Table 1-3 ESA CanSat constraints adopted as assumptions

Constraint (ESA CanSat)	Value	STRATOS Rev A
Envelope	Ø66 mm × 115 mm (parachute excluded; ≈45 mm extra for parachute, fittings, antennas)	Ø65 × 113 mm rigid; parachute and bridle outside
Mass	300–350 g; ballast if lighter	≈243 g estimated → ≈82 g ballast for a 325 g target
Descent rate	8–11 m/s recommended; never < 5 or > 12 m/s	≈9 m/s at 325 g (calculated)
Survive acceleration	20 g	Retention by screws, zip ties, rods; verify by test (IMU range ±16 g saturates above 16 g)
Operating time	4 h continuous, battery and master switch easy to reach	≈43 h estimated; switch reachable through cap; cell needs shell removal (≈2 min)
Parachute connection	50 N (ESA 2022–23 [S3]); 500 N (CanSat Belgium 2026–27 [S6])	Designed for a 500 N proof load (§6.7)
Primary mission	Temperature and pressure to the ground at ≥ 1 Hz	T and P in every 1 Hz packet; 10 Hz onboard
Budget	≤ €500 for the flight unit	Payload production ≈₹11,053 (INR, see Chapter 7)

Constraint (ESA CanSat)	Value	STRATOS Rev A
Prohibited	Explosives, pyrotechnics, flammable or dangerous materials	None used; Li-ion cell handled per §9.3

CanSat India (IN-SPACe/ISRO/ASI) uses a different class of payload (≈1 kg, 7U, rocket to ≈1 km) and is not used as the baseline [S36].

Radio regulation in India [S7, S8]

The 865–868 MHz band is licence-exempt for short-range devices under **GSR 853(E), 10 Dec 2021**. For tracking, tracing and data-acquisition devices (Table II, whose notes list meteorological and pollution sensors) the limits are **500 mW ERP, ≤ 200 kHz bandwidth, ≤ 2.5 % duty cycle for non-access-point devices, adaptive power control required**. (Other licence-exempt bands exist — 2.4 GHz [S9] and 433.05–434.79 MHz [S10] — but the selected Ra-01H covers 803–930 MHz only.) Table I (non-specific devices) allows 25 mW ERP with 1 % duty and frequency hopping. The rules state that equipment shall be type approved (ETA).

Consequence for the design: LoRa SF7 at 1 packet/s would occupy the channel 9.2 % of the time — above the limit. The baseline therefore uses the same SX1276 in **GFSK** mode: 11.46 ms per packet (1.15 % duty), ≈118 kHz occupied bandwidth and ≈10 mW ERP (with the assumed −2 dBi payload antenna). Two compliance questions remain open and are logged as risk R14: (a) adaptive power control on a one-way link — STRATOS sets the lowest power that gives an adequate margin in the range test, but has no closed loop; (b) whether a one-off student prototype built from a module needs its own ETA. The team should confirm both with the institution before any outdoor transmission.

Testing, drops and flight [S11, S12, S13]

Drone Rules 2021 do not expressly cover dropping payloads; the Aircraft Rules 1937 (Rule 26) prohibit dropping articles or parachute descents from aircraft — which include balloons and kites — without a written government order. Model-rocket propellant is treated as an explosive (PESO) [S37]. **Plan: ground and fixed-structure drop tests only until written permissions exist.**

1.4 Intended operating conditions (ASSUMED)

Table 1-4 Assumed operating conditions

Parameter	Assumed value	Basis
Deployment	Release at ≤1 km above ground (rocket, drone or balloon), then parachute descent	ESA CanSat mission profile [S5]
Ambient temperature	0 to +45 °C at ground; ≈6.5 °C/km lapse	Central India; BME280 full accuracy 0–65 °C [S20]
Pressure	≈850–1013 hPa	Ground elevations up to ≈1.5 km
Logging window	4 h per session, several sessions per day	ESA powered-time requirement
Radio range	≤ 1 km slant range (planning), 3 km stretch	Link budget §6.6
Shock	Landing at ≤ 11 m/s; 20 g survival	ESA requirement

1.5 Functional requirements

Table 1-5 Requirements (each is verified in Table 8-1)

ID	Requirement	Rationale
FR-01	Measure air temperature, relative humidity and pressure	Mission objective 1
FR-02	Measure 3-axis acceleration and angular rate	Mission objective 1 (motion)
FR-03	Record every 100 ms with session, rec_seq, t_ms and status	Mission objective 2
FR-04	Store every record onboard independently of telemetry	Mission objective 3
FR-05	Detect incomplete or corrupted onboard lines	Mission objective 3
FR-06	Transmit one 44-byte packet per second with pkt_seq, rec_seq and CRC-16	Mission objective 4
FR-07	Ground station validates frames and logs accepted and rejected frames with UTC time	Mission objective 7
FR-08	Ground station detects missing, duplicate and out-of-order packets and session changes	Mission objective 5
FR-09	Reconcile onboard and ground records and classify every discrepancy	Mission objective 5
FR-10	Distinguish an unavailable measurement from a valid zero	Project brief §4.3
FR-11	Operate for the design logging window of 4 h on one charge with ≥ 50 % margin	Assumed operating requirement

ID	Requirement	Rationale
FR-12	Fit the reference envelope (Ø 66 × 115 mm) and the competition mass limit	Assumed envelope (rules TBC)
FR-13	Descend under a parachute within the target descent-rate band	Assumed (rules TBC)
FR-14	Provide an audible recovery aid after landing	Recovery objective
FR-15	Keep radio emissions within the India licence-exempt 865–867 MHz limits	Regulatory
FR-16	Retain all components under handling, drop and landing loads	Mission objective 6
FR-17	Protect the Li-ion cell against short circuit and over-discharge	Safety
FR-18	Close the log cleanly before battery cut-off	Mission objective 3

1.6 Constraints and success criteria

- **Constraints:** student build with ordinary tools (3D printer, soldering, hand tools); parts available in India; reference envelope Ø66 × 115 mm; licence-exempt radio limits; no flight without permission; budget within a student team's means.
- **Success (prototype stage):** a bench prototype that logs 4 h of real data, transmits it at 1 Hz, and whose reconciliation explains every gap with zero unexplained discrepancies.
- **Success (mission stage, if authorised):** a permitted drop or flight in which the descent rate is within 8–11 m/s, the payload is recovered intact, and the onboard and ground records reconcile.

2 System architecture

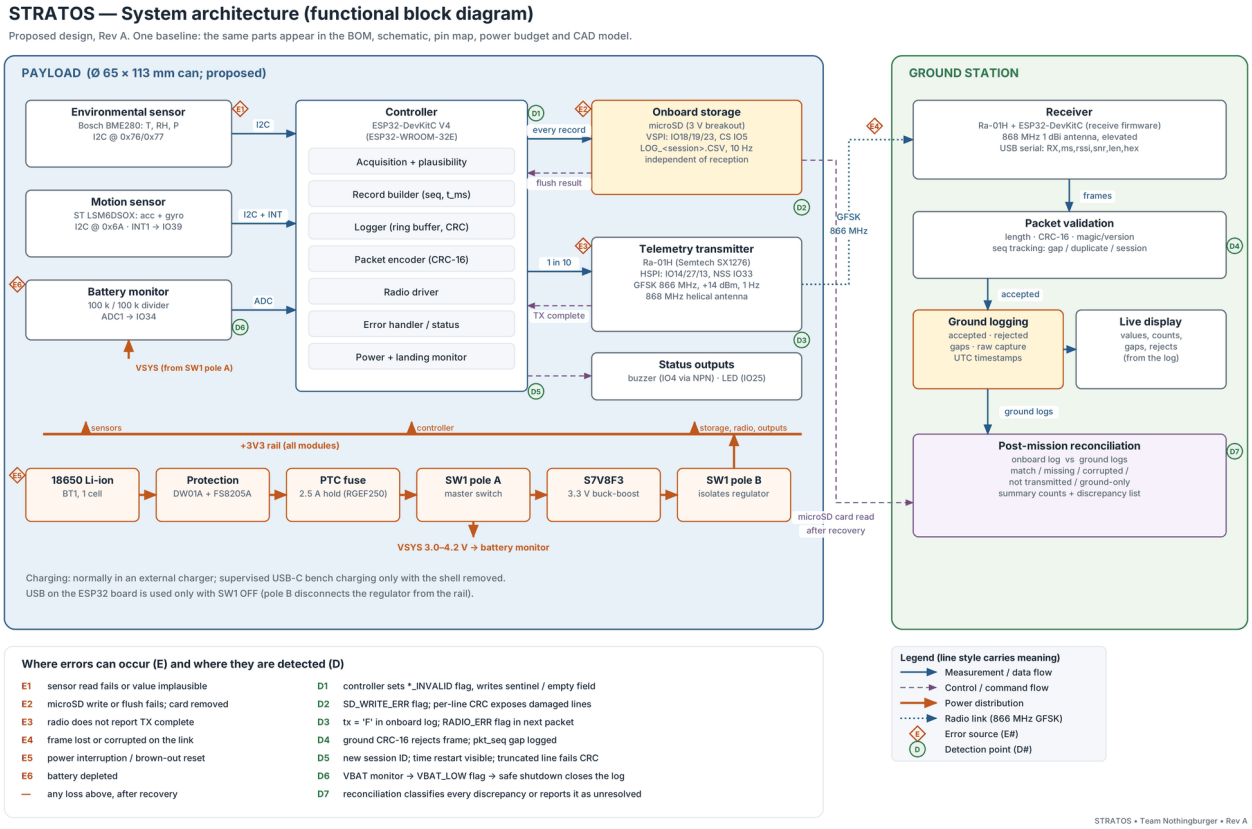


Figure 2-1 Functional block diagram. Line style distinguishes data, control, power and radio flows; E# marks where errors originate and D# where they are detected.

2.1 Subsystems

Table 2-1 Subsystems and interfaces

Subsystem	Function	Key parts	Interface to controller
Environmental sensing	T, RH, P at 10 Hz	BME280 breakout	I2C0 (IO21/IO22), 400 kHz
Motion sensing	3-axis acceleration and rate	LSM6DSOX (Adafruit 4438)	I2C0 + INT1 → IO39
Controller	Scheduling, records, packets, error handling	ESP32-DevKitC V4	—
Onboard storage	Every record, CSV + line CRC	microSD 3 V breakout + 16 GB card	VSPI (IO18/19/23, CS IO5)
Telemetry	1 Hz packets, GFSK 866 MHz	Ra-01H (SX1276) + helical antenna	HSPI via GPIO matrix (IO14/27/13, NSS IO33), RST IO32, DIO0 IO26
Power	Protected, switched, regulated 3.3 V	18650, DW01A module, PTC, DPDT, S7V8F3	VSYS sense → IO34 (ADC1)
Indicators	Status and recovery beeper	LED, buzzer via BC547	IO25, IO4
Structure and recovery	Protect, carry loads, descend safely	PETG caps/spine/sleeve, M3 rods, Al yoke, parachute	—
Ground station	Receive, validate, log, display, reconcile	Ra-01H + ESP32 receiver, laptop, stratos_gs	USB serial 115 200 baud

2.2 Data flow and the three data objects

Every 100 ms the controller builds one **record** (session, rec_seq, elapsed time, 10 measurement fields, status flags). Every record goes to the onboard log; every tenth record (rec_seq mod 10 = 1) is also encoded as a 44-byte **telemetry packet** with its own pkt_seq and CRC. The ground station turns each valid packet into a **received record** with a UTC reception time, RSSI and sequence event. These three objects are deliberately different and are never confused in the software or the documents.

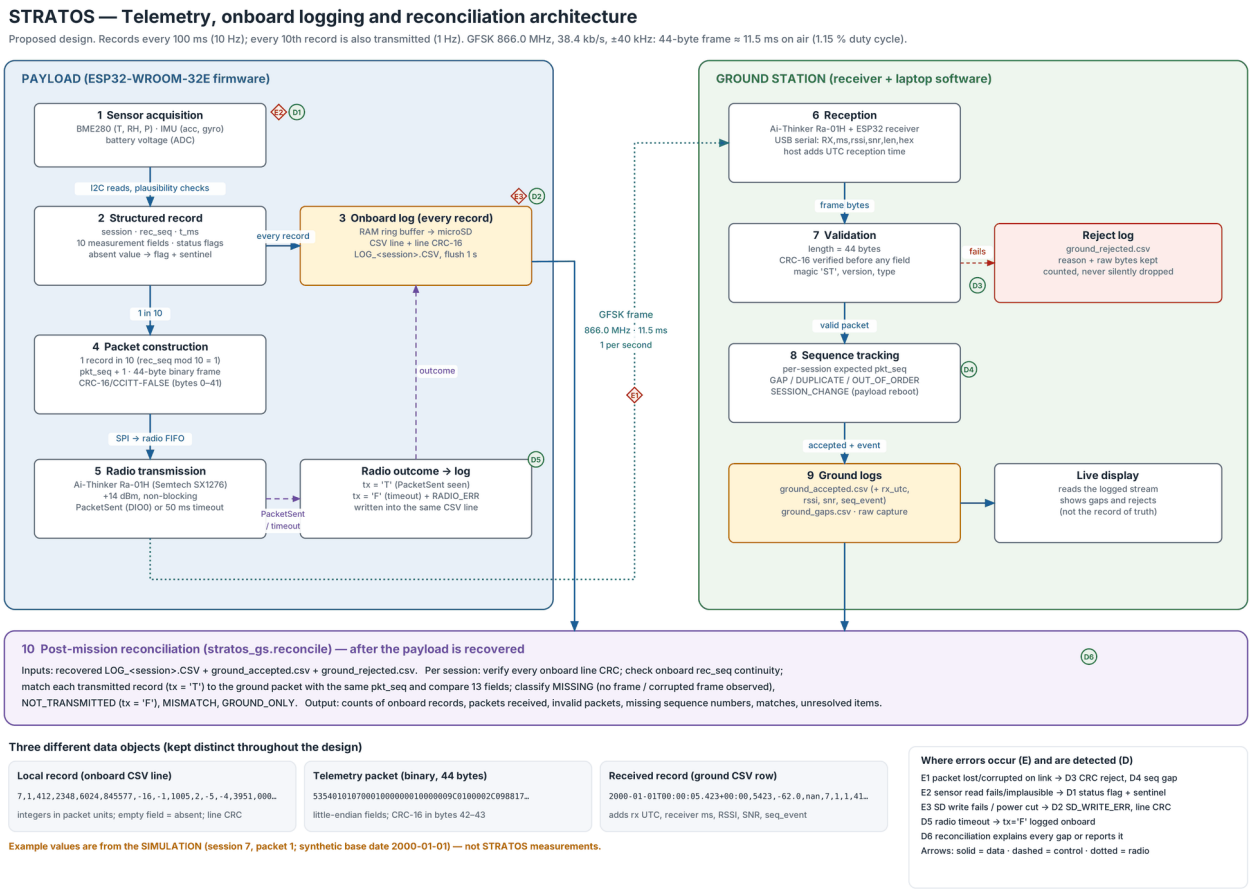


Figure 2-2 Telemetry, onboard logging and reconciliation architecture. Example values are simulated.

2.3 Power flow

Cell (BT1) → DW01A/FS8205A protection (U7) → PTC fuse F1 → master switch pole A → VSYS (3.0–4.2 V) → S7V8F3 buck-boost (U6) → master switch pole B → +3V3 rail → all modules. Pole B isolates the regulator when the switch is OFF so that the ESP32 board's USB port can power it for programming: Espressif warns that only one of the board's supply options may be used at a time [S16].

2.4 Key design decisions

Table 2-2 Decisions, alternatives and reasons

Decision	Alternatives considered	Reason
ESP32-DevKitC V4 controller	Arduino Nano/Uno; RP2040 Pico; ESP32-S3 DevKit	Enough GPIO for two SPI buses + I2C + ADC1; 520 kB SRAM; mature libraries; original board in stock (Mouser); fits on a 50 × 70 mm carrier
GFSK instead of LoRa at 1 Hz	LoRa SF7 at 1 Hz; LoRa at 0.25 Hz; 2.4 GHz	GSR 853(E) duty-cycle limit 2.5 %: LoRa SF7 needs 92 ms per packet; GFSK needs 11.5 ms. LoRa kept as a slower, longer-range option
Two separate SPI buses	SD and radio sharing one bus	Avoids SD cards that do not release MISO; keeps radio timing independent of SD writes
CSV log with per-line CRC	Binary log; plain CSV	Human-readable, easy to reconcile, and truncated/corrupted lines are detectable
Integers in packet units in the log	Floating-point CSV	Exact field-by-field comparison with packets, no rounding ambiguity
Absent value = flag + sentinel/empty field	Write 0	A missing reading can never be mistaken for a valid zero
Particulate sensor excluded	PMS5003	Needs 5 V and ≤100 mA for its fan, 50 × 38 × 21 mm, 30 s warm-up, airflow path [S31]; no mission objective requires it
GNSS optional	NEO-M8N fitted	Not needed for the data-integrity mission; adds ≈30 mA [S32]; UART2 pins reserved for recovery aid if rules require it
External cell charging	On-board USB charging	No charging inside a closed can; USB-C on U7 only for supervised bench use

Decision	Alternatives considered	Reason
Rod-and-spine structure	Stacked decks; tube with glued boards	

3 Mechanical design

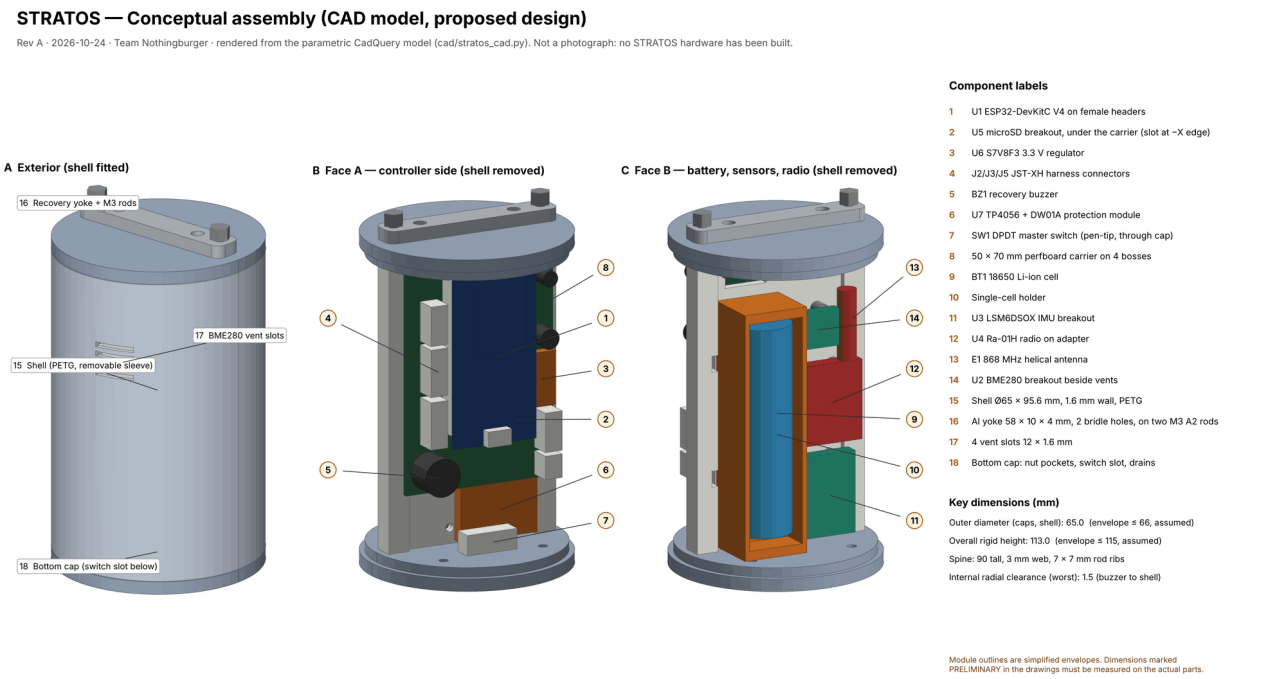


Figure 3-1 Conceptual assembly rendered from the parametric CAD model (not a photograph). Numbered labels are listed on the figure.

3.1 Arrangement

A 3 mm PETG **spine** with two 7 × 7 mm edge ribs runs between a bottom and a top cap. Two **M3 A2 stainless threaded rods** pass through the ribs; nuts in hex pockets under the bottom cap and nyloc nuts above an **aluminium yoke** clamp the stack. Face A (+Y) carries the controller on a 50 × 70 mm perfboard **carrier**; face B (−Y) carries the 18650 cell, IMU, radio and BME280. The **shell** (Ø65 × 1.6 mm wall, 95.6 mm long) is a non-structural sleeve located by the cap spigots.

Table 3-1 Main dimensions (mm)

Feature	Value	Status
Cap and shell outer diameter	65.0	PROPOSED (envelope ≤ 66, ASSUMED)
Overall rigid height (bottom face to rod tops)	113.0	PROPOSED (envelope ≤ 115, ASSUMED)
Cap flange / spigot	4 / 3 (spigot Ø61.6)	PROPOSED
Shell bore	Ø61.8	PROPOSED
Spine height × width	90 × 57	PROPOSED
Rod centres	50	PROPOSED
Recovery yoke	58 × 10 × 4, bridle holes at ±16	PROPOSED
Worst internal radial clearance	1.54 (BUZZER to shell bore)	CAD check
Interferences between modelled parts	0	CAD check (module outlines PRELIMINARY)

3.2 Mounting, access and cable routing

- Modules are screwed (M2.5 self-tapping) to printed bosses; the cell is held by its holder **and** a zip tie through the spine.

- The microSD breakout sits under the carrier, card edge at −X; the ESP32 micro-USB faces −Z. Both are reached by removing the shell.
- Service sequence: undo two nyloc nuts → lift yoke and top cap → slide the shell upwards. Estimated 2 minutes (to be timed).
- SW1 (DPDT) is operated with a pen tip through an 8 × 3 mm slot in the bottom cap, which prevents accidental switching.
- Harnesses (JST-XH) cross between faces through two spine slots (14 × 6 and 10 × 5 mm); vent slots in the shell sit beside the BME280.
- The antenna is a helix soldered to the radio adapter at the top of face B, ≥ 30 mm from the cell and on the opposite face from the ESP32.

3.3 Materials and fabrication

Table 3-2 Materials and fabrication

Part	Material	Process	Notes
Caps, spine, shell	PETG	FDM print, 0.4 mm nozzle	Spine printed upright (rod bores vertical); caps flange-down; shell 3 perimeters
Rods	M3 A2 stainless threaded rod	Cut to 112.5 mm, deburr	Threadlocker or nyloc nuts
Yoke	6061-T6 flat bar 10 × 4 mm	Saw, file R3 corners, drill	Break edges so the bridle is not cut
Carrier	Double-sided perfboard 50 × 70 mm	Hand wiring	A custom PCB is a later option
Parachute	Ripstop nylon, paracord inner strands	Cut and sew; spill hole optional	High-visibility colour preferred

3.4 Recovery concept

A Ø0.33 m flat circular canopy hangs from a swivel and a two-leg (Y) bridle attached to holes at ±16 mm on the yoke, close to the rods so the yoke is not loaded in large bending. Deployment hardware (ejection from a rocket, release from a platform) is outside the Rev A scope and depends on the competition's launch provider.

3.5 Unresolved mechanical items

Table 3-3 Unresolved mechanical items

#	Item	Status
A1	Envelope and mass limits are ESA CanSat values, not SEVA FIRST rules	ASSUMED
A2	Rigid height excludes the parachute, Y-bridle and swivel	ASSUMED
A3	PETG printed with a 0.4 mm nozzle; spigot fit 0.1 mm radial	PROPOSED
A4	Breakout outlines (BME280, adapter, holder, switch, connectors) from listings, not drawings	PRELIMINARY
A5	JST-XH mated height at the carrier edge leaves ≈2.4 mm to the bore	TO VERIFY with real parts
A6	Shell anti-rotation feature not yet designed	UNRESOLVED
A7	Ballast location for ≈82 g (if a 300 g minimum applies)	UNRESOLVED
—	Holder and cell retention under 20 g; PETG cap strength under the rod nuts	To be proven by test (FR-16)

The four drawing sheets (general arrangement, internal arrangement with section, exploded view with parts list, part details) are in STRATOS_Mechanical_Drawings.pdf and reproduced in Appendix F.

4 Electronics

4.1 Component selection

Table 4-1 Selected parts and verified key data

Ref	Part	Key data (source)	Why chosen
U1	ESP32-DevKitC V4 (WROOM-32E)	3.0–3.6 V; modem-sleep 80 MHz 20–31 mA; board 54.4 × 27.9 mm; 38 pins [S14–S16]	Two SPI hosts, ADC1, enough GPIO; original stocked
U2	BME280 breakout	1.71–3.6 V; ±1 hPa, ±3 %RH, ±0.5 °C at 25 °C; 3.6 µA at 1 Hz; I2C 0x76/0x77 [S20]	T, RH and P in one part; low power
U3	LSM6DSOX (Adafruit 4438)	0.55 mA accel+gyro HP; ±2–16 g, ±125–2000 dps; 25.6 × 17.8 mm, 1.7 g; I2C 0x6A [S21, S22]	Active part (MPU-6050 is EOL [S23]); level-shifted breakout
U4	Ra-01H (SX1276)	803–930 MHz; 17 × 16 × 3.2 mm; stamp pads, soldered antenna; TX 87 mA at +17 dBm; standby 1.6 mA [S24, S25]	868 MHz-band part in stock in India; FSK and LoRa in one chip
U5	microSD 3 V breakout (Adafruit 4682)	No regulator or level shifter (3 V logic only); 25.4 × 22.8 × 3.5 mm [S27]; SD ≤100 mA in SPI [S26]	Direct 3.3 V SPI, no 5 V-only translator
U6	Pololu S7V8F3	VIN 2.7–11.8 V; 3.3 V out; up to 1 A (VIN > 3.3 V); Iq < 0.1 mA; 11 × 17 mm [S28]	Keeps 3.3 V over the whole Li-ion range
U7	TP4056 module with DW01A/FS8205A	TP4056: 4.2 V ±1.5 %, ≤1 A [S29]; DW01A thresholds NOT CONFIRMED	Cheap, available cell protection
BT1	18650 Li-ion — BOM candidate Molicel INR18650-P28A	2,800 mAh per supplier listing (datasheet to be checked). Envelope and mass upper bound from the Samsung 35E datasheet: ≥3,350 mAh, 8 A cont., ≤50 g, Ø18.55 × 65.25 mm [S30]	Robust, swappable, counts toward minimum mass

Excluded or deferred. PMS5003 particulate sensor (5 V, ≤100 mA fan, 50 × 38 × 21 mm, 30 s warm-up) — excluded; NEO-M8N GNSS (23–32 mA tracking, 16 × 12.2 mm) — optional, UART2 pins reserved; ICM-42688-P — no breakout found in India; LSM6DS3TR-C — 0.90 mA, acceptable fallback.

> Datasheet discrepancies to resolve before ordering: BME280 temperature accuracy over 0–65 °C (±0.5 °C in rev 1.24 vs ±1.0 °C in rev 1.6); Ra-01H maximum output (+19 vs +20 dBm in the same document). SX1276 GFSK sensitivity at 38.4 kb/s was **not confirmed** and is an assumption in the link budget.

4.2 Schematic

The schematic is generated from a single netlist (design/netlist.py) that also produces the pin map, the component list and a KiCad-format netlist, so the four cannot disagree. Net labels of the same name are connected. Sheet 1 is reproduced below; all three sheets are in STRATOS_Electronics_Schematic.pdf and Appendix F.

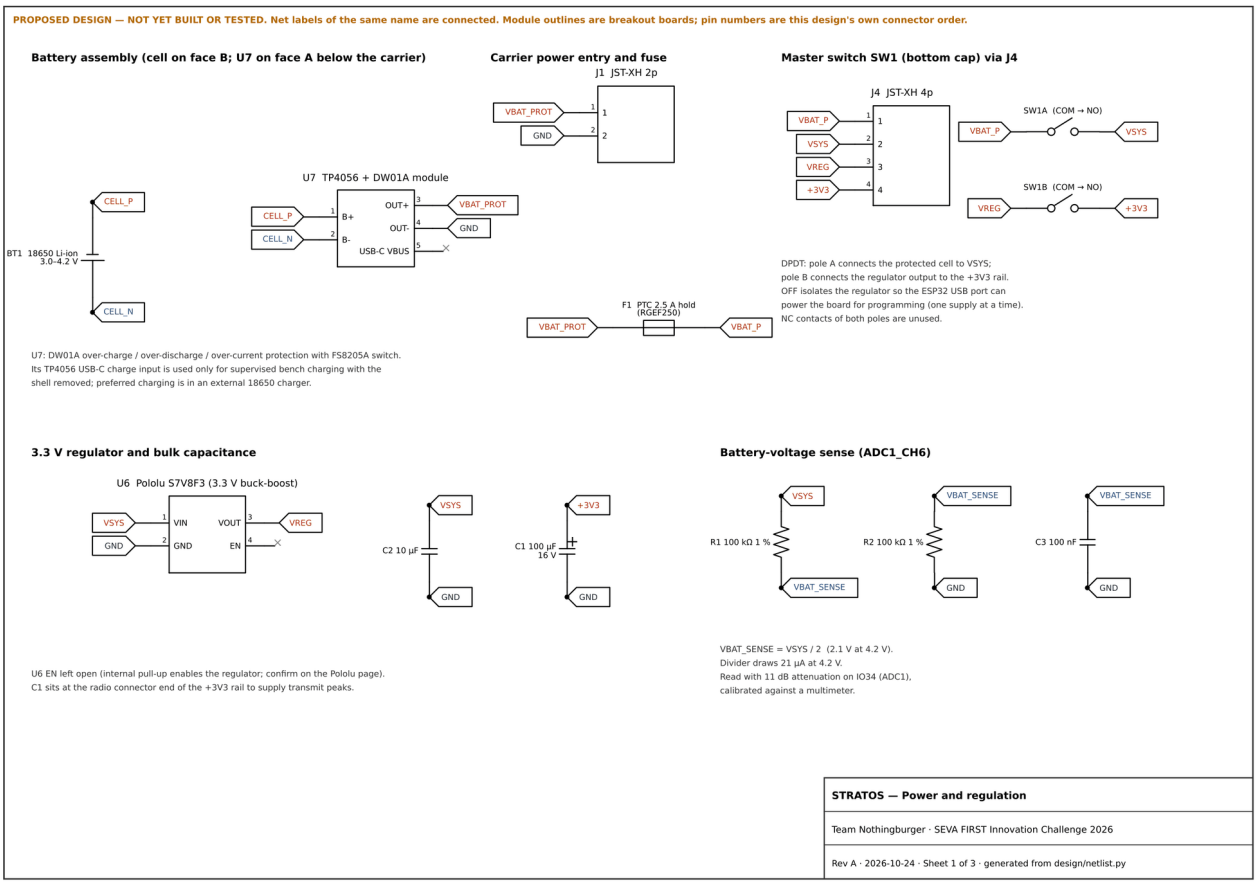


Figure 4-1 Schematic sheet 1 of 3 — power, protection, master switch, regulator and battery sense.

4.3 Pin mapping

Table 4-2 ESP32 pin assignment (full table with far-end pins in STRATOS_Pin_Mapping.csv)

Signal	GPIO	Dir.	Bus	Connects to	Note
I2C_SDA	21	Bi	I2C0	U2/U3	400 kHz; pull-ups on breakout boards (check total value)
I2C_SCL	22	Out	I2C0	U2/U3	400 kHz
IMU_INT1	39	In	GPIO	U3	IMU data-ready interrupt; input-only pin, no internal pull
SD_SCK	18	Out	VSPI	U5	native VSPI pin
SD_MISO	19	In	VSPI	U5	native VSPI pin
SD_MOSI	23	Out	VSPI	U5	native VSPI pin
SD_CS	5	Out	VSPI	U5	strapping pin, must be HIGH at boot: R4 10 kΩ pull-up holds card deselected
RADIO_SCK	14	Out	HSPI	U4	native HSPI CLK; toggles at boot while NSS is held high (harmless)
RADIO_MISO	27	In	HSPI	U4	GPIO matrix (avoids strapping pin IO12)
RADIO_MOSI	13	Out	HSPI	U4	native HSPI MOSI
RADIO_NSS	33	Out	HSPI	U4	chip select; R5 10 kΩ pull-up keeps radio deselected at boot
RADIO_RST	32	Out	GPIO	U4	active-low reset, used for radio recovery
RADIO_DIO0	26	In	GPIO	U4	PacketSent / PayloadReady interrupt (FSK); TxDone/RxDone (LoRa)
RADIO_DIO1	35	In	GPIO	U4	FIFO level / spare; input-only pin
VBAT_SENSE	34	In	ADC1_CH6	R1/R2	VSYS via 100k/100k divider + 100 nF; ADC1 so Wi-Fi state does not matter
BUZZER_EN	4	Out	GPIO	Q1	drives Q1 base through R6 1 kΩ
STATUS_LED	25	Out	GPIO	D2	LED via R7 1 kΩ (≈1 mA)

Avoided pins: GPIO6–11 (module flash), GPIO0 and GPIO2 (boot), GPIO12 (must be low at power-on — it selects the flash voltage) and GPIO15 (boot log). GPIO5 is a strapping pin whose default is high; R4 (10 kΩ) keeps it high and the card deselected at boot [S14, S15, S17]. Routing the radio's SPI through the GPIO matrix limits the clock to ≈26 MHz full-duplex [S18], well above the SX1276's 10 MHz maximum.

4.4 Compatibility checks

Table 4-3 Electrical compatibility checks

Check	Result	Basis
Supply voltages on +3V3 (3.3 V +5/−3 %)	Within ESP32 3.0–3.6, BME280 1.71–3.6, LSM6DSOX 1.71–3.6 (board VIN 3–5), SX1276 1.8–3.7, Ra-01H 2.7–3.6, SD 2.7–3.6	S14, S20, S22, S24–S26, S28
Logic levels	All 3.3 V CMOS; no 5 V device on the bus	U5 is a 3 V-only breakout
I2C addresses	BME280 0x76/0x77 vs LSM6DSOX 0x6A — no conflict	S20, S22
I2C pull-ups	Both breakouts include pull-ups (Adafruit: 10 kΩ); R8/R9 left unfitted unless measured too weak	S22
ADC range	VBAT/2 = 1.5–2.1 V inside the 150–2450 mV range at 11 dB	S19
Regulator capacity	Worst coincident peak 316 mA vs up to 1 A (typ. 0.5–1 A)	S28; §6.1
Module supply requirement	WROOM-32E asks for a supply able to deliver ≥ 0.5 A; Ra-01H asks for > 200 mA	S14, S25
Switch rating	DPDT rating not stated by the supplier — must be ≥ 0.5 A DC	OPEN
Strapping pins	IO5 held high (R4); IO12/IO15/IO0/IO2 unused	S14

5 Firmware and telemetry

5.1 Firmware architecture

The firmware runs a fixed 100 ms cycle as an explicit state machine. Modules: acquisition and plausibility checks; record builder (session, rec_seq, millis); logger (50-record RAM ring buffer, CSV formatting with line CRC at flush time); packet encoder (shared C code, host-tested); radio driver (non-blocking transmit, completion interrupt on DIO0, 50 ms timeout); error handler (flags, retries, recovery states); power and landing monitor. The draft sketch is in firmware/payload/ and is **not yet compiled**.

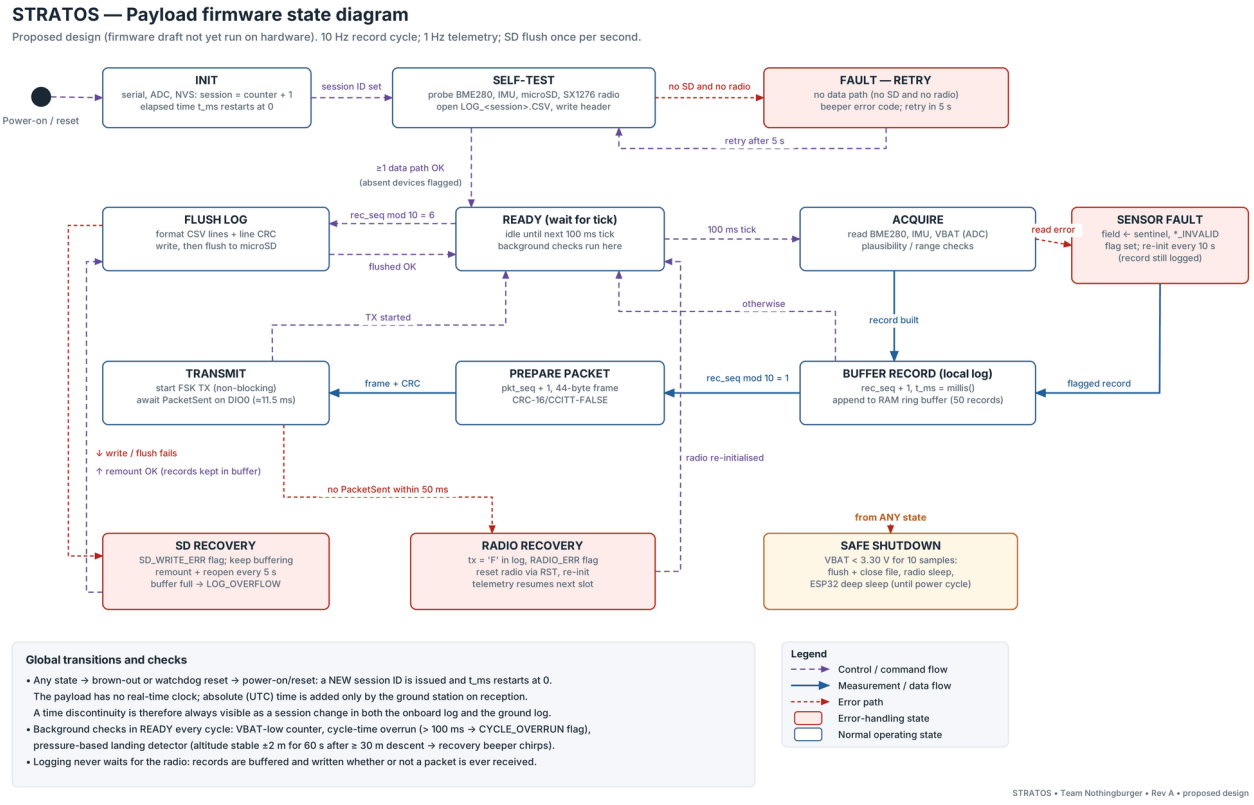


Figure 5-1 Firmware state diagram with error handling (proposed).

5.2 Sampling and logging

- All sensors are read once per 100 ms cycle into one record (BME280 normal mode ≈13 Hz output; IMU 104 Hz output data rate sampled at 10 Hz).
- Records with rec_seq mod 10 = 1 are transmitted (1 Hz); the SD card is flushed at rec_seq mod 10 = 6, half a second away from the radio slot, to avoid coincident current peaks.
- Records are kept as binary structures until flush so that the radio outcome ('T' sent, 'F' failed) is written into the same CSV line.
- A buffer overflow or write failure is never silent: LOG_OVERFLOW / SD_WRITE_ERR flags travel in the next packet and the reconciliation reports the onboard gap.

5.3 Packet format

Table 5-1 44-byte telemetry packet (little-endian); full specification in STRATOS_Packet_Format.md

Offset	Size	Field	Type	Unit
0	2	magic	char[2]	-
2	1	version	uint8	-
3	1	type	uint8	-
4	2	session	uint16	-
6	4	pkt_seq	uint32	-
10	4	rec_seq	uint32	-
14	4	t_ms	uint32	ms
18	2	temp_cC	int16	0.01 degC
20	2	rh_cpct	uint16	0.01 %RH
22	4	press_dPa	uint32	0.1 Pa
26	2	ax_mg	int16	mg
28	2	ay_mg	int16	mg
30	2	az_mg	int16	mg
32	2	gx_ddps	int16	0.1 deg/s
34	2	gy_ddps	int16	0.1 deg/s
36	2	gz_ddps	int16	0.1 deg/s
38	2	vbat_mV	uint16	mV
40	2	status	uint16	bitfield
42	2	crc16	uint16	-

Integrity field: CRC-16/CCITT-FALSE (polynomial 0x1021, initial 0xFFFF, no reflection) over bytes 0–41; check value 0x29B1 for '123456789'. The receiver verifies it before reading any field. The radio's own packet CRC is also enabled. **Absent values** are flagged in the status word and written as sentinels (packet) or empty fields (log).

Table 5-2 Status flags

Bit	Flag	Meaning
0	BME_PRESENT	BME280 initialised and responding
1	IMU_PRESENT	IMU initialised and responding
2	SD_OK	Onboard log file open and last flush succeeded
3	T_INVALID	Temperature unavailable or out of plausibility range
4	RH_INVALID	Humidity unavailable or out of plausibility range
5	P_INVALID	Pressure unavailable or out of plausibility range
6	ACC_INVALID	Acceleration unavailable
7	GYR_INVALID	Angular rate unavailable
8	SD_WRITE_ERR	At least one SD write/flush failed since last packet
9	LOG_OVERFLOW	RAM log buffer overflowed; records were dropped
10	VBAT_LOW	Battery voltage below 3.40 V
11	RADIO_ERR	Previous transmission failed or timed out
12	CYCLE_OVERRUN	Acquisition cycle exceeded its 100 ms period

5.4 Ground-station processing

1. Receiver firmware forwards every frame as `RX,<ms>,<rssi>,<snr|nan>,<len>,<hex>`; the laptop stamps UTC time and saves the raw capture.
2. Validation: length = 44 → CRC-16 → magic/version/type. Failures go to ground_rejected.csv with the raw bytes and a reason.
3. Sequence tracking per session: GAP (missing numbers logged to ground_gaps.csv), DUPLICATE, OUT_OF_ORDER, SESSION_CHANGE.
4. Accepted packets go to ground_accepted.csv; the display reads the logged stream, so what is shown is what was recorded.

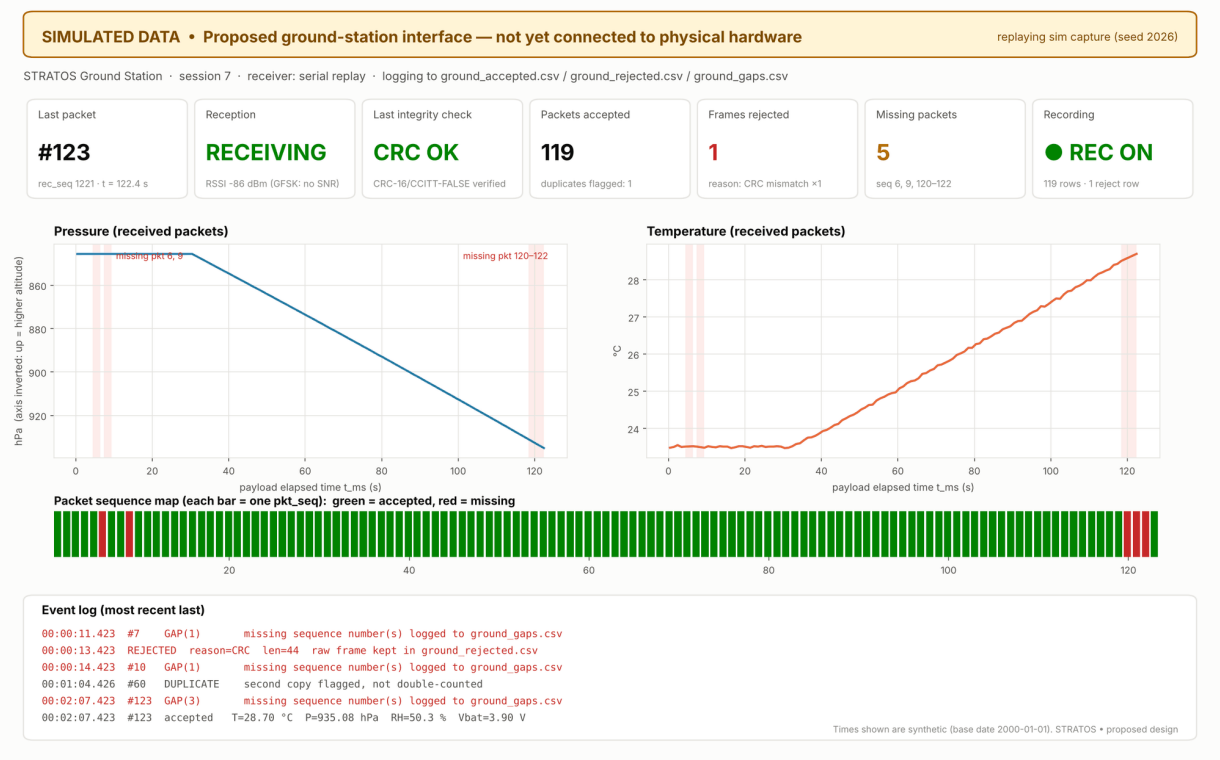


Figure 5-2 Proposed ground-station interface drawn from the prototype software's state while replaying a SIMULATED capture — not connected to hardware.

5.5 Post-mission reconciliation rules

Table 5-3 Reconciliation rules

Rule	Applies to	Classification
R1	Every onboard line	Line CRC fails → ONBOARD_LINE_INVALID
R2	Onboard rec_seq jumps	ONBOARD_LOGGING_GAP
R3–R5	Transmitted (tx = 'T') and received	13 fields compared → MATCH or MISMATCH
R6	Transmitted, not received	MISSING — corrupted frame observed / no frame observed
R7	tx = 'F' onboard	NOT_TRANSMITTED (explained)
R8	Received, no valid onboard record	GROUND_COPY_OF_INVALID_ONBOARD_LINE (explained) or GROUND_ONLY (unresolved)

5.6 Software demonstration (SIMULATED)

stratos_gs.simulate generates a synthetic 171 s session (hold at 1 km, 9 m/s descent, landing), writes it in the onboard format, passes the packets through a simulated link with **deliberately injected faults**, runs the ground tracker and then the reconciliation:

Table 5-4 Injected faults and results (simulated)

Injected fault (synthetic)	Detected by	Reconciliation result
Packet 6 dropped	Ground gap log	MISSING — no frame observed; recovered from onboard log
One bit flipped in packet 9	CRC-16 reject	MISSING — corrupted frame observed; recovered from onboard log
Packets 120–122 lost (burst)	Ground gap log	MISSING — no frame observed (3)
Radio failure on packet 150	Onboard tx = 'F', RADIO_ERR	NOT_TRANSMITTED (explained)
Packet 60 delivered twice	DUPLICATE flag	Counted once
Humidity unavailable, records 800–804	RH_INVALID flag, empty field	Absent ≠ zero
Final log line truncated	Line CRC	Ground packet preserves that record (explained)

Result (synthetic): 1,710 valid onboard lines plus 1 truncated final line; 171 packets scheduled in the valid lines (170 sent, 1 radio failure) plus packet 172 in the truncated line; 166 unique packets accepted at the ground (the duplicate counted once) and 1 corrupted frame rejected; 5 sequence numbers missing (6, 9, 120–122), all recovered from the onboard log (the ground gap log also lists 150, which the onboard log shows was never transmitted); 165 received packets match their onboard records field for field, 0 mismatch, and packet 172 preserves the record whose onboard line was truncated. **Unresolved discrepancies: 0.**

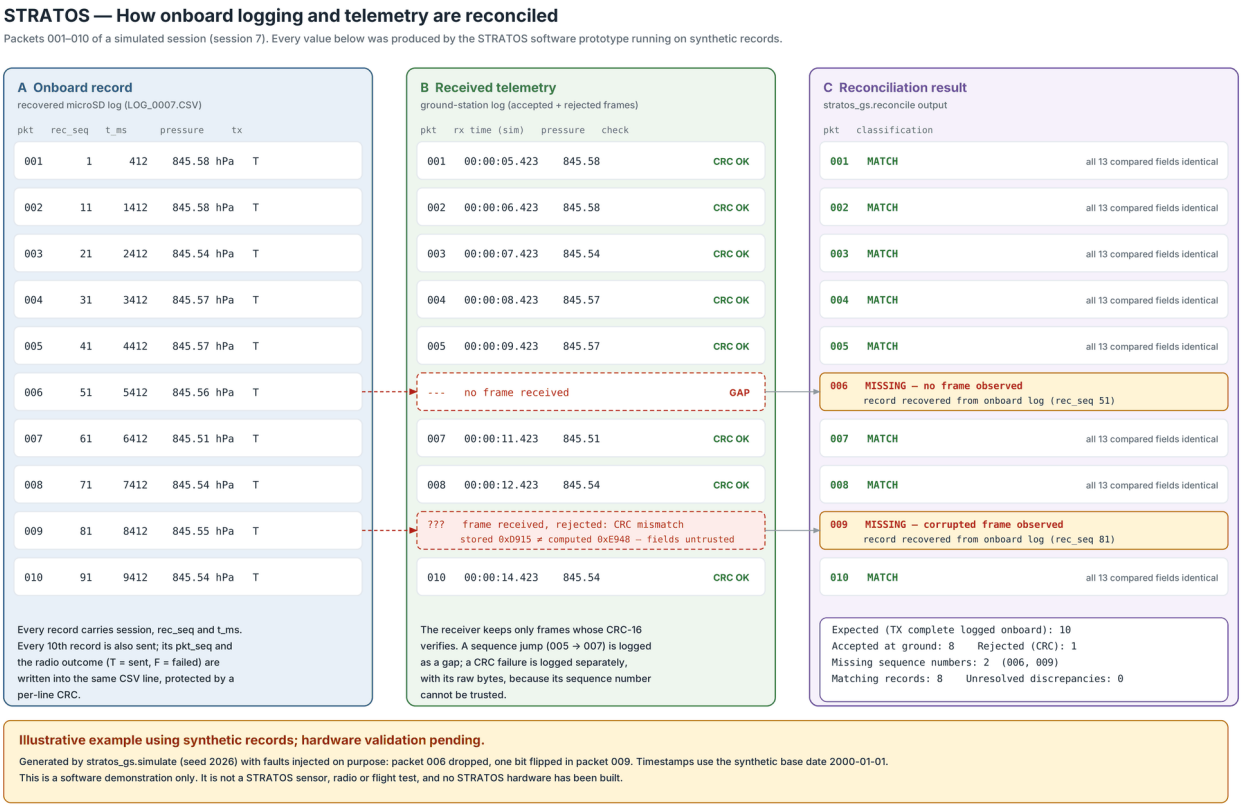


Figure 5-3 Data-integrity demonstration, packets 001–010. Illustrative example using synthetic records; hardware validation pending.

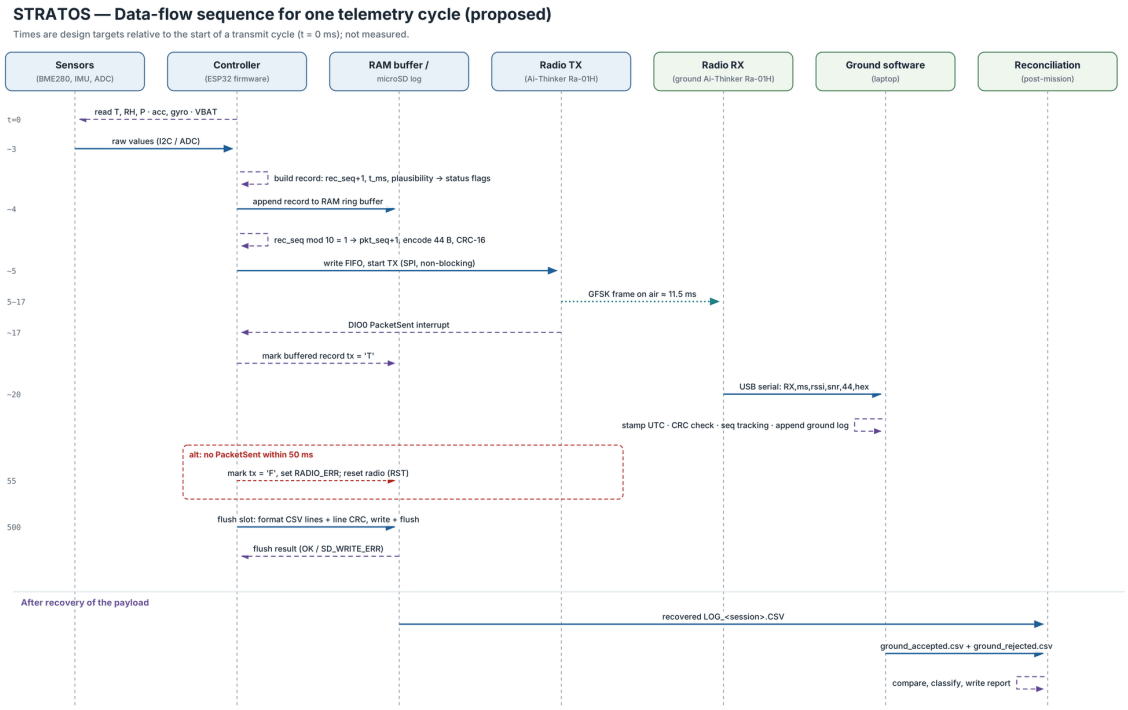


Figure 5-4 Data-flow sequence for one telemetry cycle (design targets, not measurements).

6 Engineering calculations

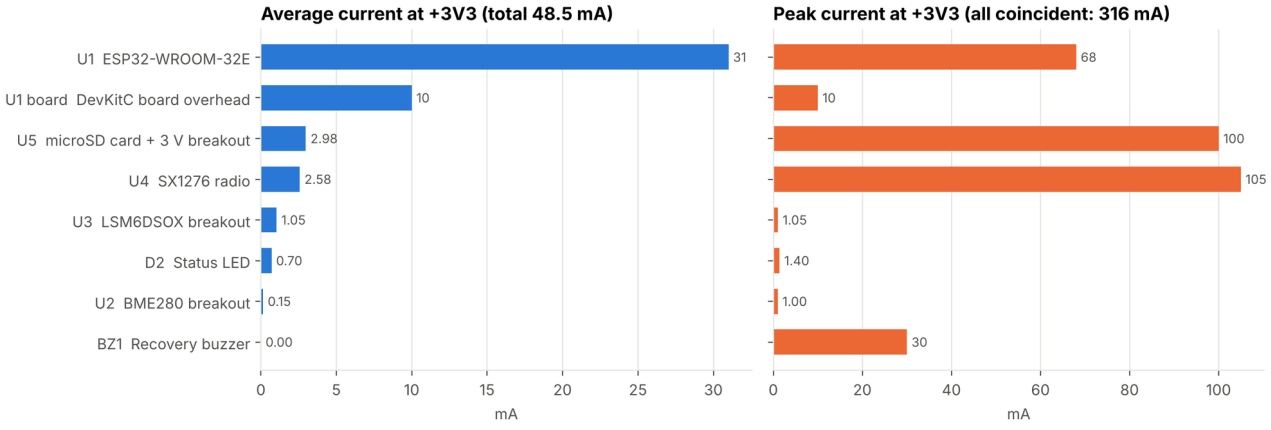
All calculations are formula-driven in the workbooks; inputs are blue and labelled VERIFIED or ASSUMED. Each result below is preliminary until the ASSUMED inputs are replaced by measurements.

6.1 Power budget

Average current per load: I_avg = I_active·D + I_idle·(1 − D). Battery current: I_batt = P_rail / (η·V_nom) + battery-side loads. Ideal operating time: t = C / I_avg.

Table 6-1 Power budget at the +3V3 rail

Load	Active mA	Idle mA	Duty	Avg mA	Peak mA	Source
ESP32-WROOM-32E, modem-sleep 80 MHz, Wi-Fi/BT off	31.00	31.00	100.00 %	31.00	68	VERIFIED
DevKitC board overhead (USB-UART, power LED, regulator back-feed)	10.00	10.00	100.00 %	10.00	10	ASSUMED
BME280 breakout (10 Hz normal mode)	1.00	0.05	10.00 %	0.15	1	ASSUMED active ≈1 mA during conversion (datasheet 3.6 µA at 1 Hz scales ≈10× at 10 Hz)
LSM6DSOX breakout, accel + gyro high-performance 104 Hz	1.05	1.05	100.00 %	1.05	1	VERIFIED 0.55 mA (DS12814) + ASSUMED 0.5 mA board regulator/pull-up overhead
microSD card + 3 V breakout (1 flush per second)	100.00	1.00	2.00 %	2.98	100	VERIFIED limit
SX1276 radio (Ra-01H), GFSK +14 dBm on PA_BOOST	87.00	1.60	1.15 %	2.58	105	VERIFIED 87 mA at +17 dBm PA_BOOST used as upper bound for +14 dBm (SX1276 Table 6)
Status LED (1 kΩ, toggles at 1 Hz)	1.40	0.00	50.00 %	0.70	1	COMPUTED (3.3 V − ≈1.9 V LED drop) / 1 kΩ
Recovery buzzer (off during flight)	30.00	0.00	0.00 %	0.00	30	ASSUMED 30 mA (5 V-rated electromagnetic buzzer; not measured at 3.3 V)
+3V3 total				48.5	316	



Values marked ASSUMED in the power budget (board overhead, SD idle/write time, buzzer) must be measured. Source: STRATOS_Power_Budget.xlsx.

Figure 6-1 Largest contributors to average and peak current.

Table 6-2 Battery current and operating time (estimates)

Result	Value
Rail power	160 mW
Battery current (η = 0.85 ASSUMED, 3.6 V)	52.4 mA
Ideal time, 2,800 mAh BOM cell (listing value)	53 h
Estimated time, 80 % usable (design basis)	43 h → margin 10.7× over 4 h
Estimated time, genuine Samsung 35E (3,350 mAh minimum)	51 h
Estimated time, conservative 2,000 mAh cell	31 h → margin 7.6× over 4 h

Result	Value
Worst coincident peak at the cell (3.0 V)	410 mA vs 8 A (35E rating; P28A rating to be read from its datasheet)

The largest uncertainty is the DevKitC board overhead (no official figure) — the first bench measurement. Wi-Fi left on by mistake would add ≈239 mA and cut the time to ≈7 h (still above 4 h). FR-11 asks that the 4 h window uses at most 50 % of the usable capacity, i.e. a margin of at least 2×.

6.2 Storage and sampling

Table 6-3 Storage and sampling figures

Quantity	Value
Record rate × window	10 records/s × 4 h = 144,000 records
Worst-case line (122 characters + CR LF)	124 bytes → 17.9 MB per session
Typical line (simulated mean)	≈70 bytes → 10.1 MB per session
Usable card capacity (16 GB × 0.93 ASSUMED)	≈14.9 GB → one session uses 0.12 %
Average write rate	≈1.2 kB/s worst case
RAM ring buffer	50 × 48 B = 2.4 kB (< 0.5 % of SRAM)

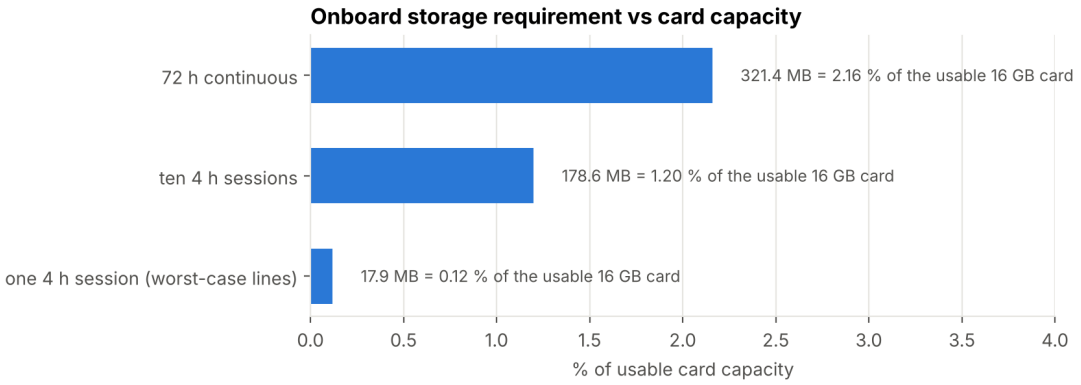
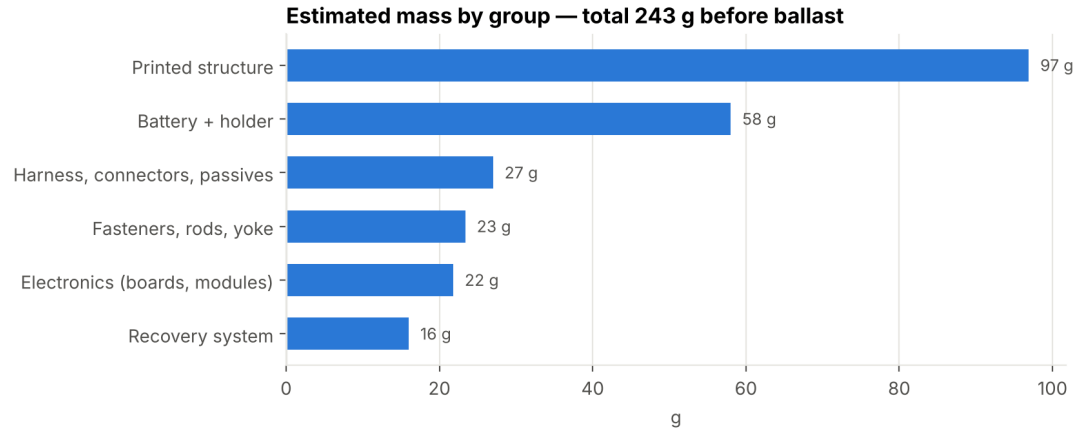


Figure 6-2 Storage requirement against card capacity.

6.3 Mass budget



Most entries are estimates (EST) to be replaced by weighing; if ESA-style rules apply, ≈82 g of ballast brings the payload to the 325 g design target.

Figure 6-3 Estimated mass by group (most items EST until weighed).

Estimated total before ballast: **243 g**. If a 300–350 g rule applies, ≈57–107 g of ballast is needed (≈82 g for the 325 g design target). Printed parts are CAD volumes × assumed fill factors.

6.4 Dimensional checks (CAD)

All modelled parts lie within Ø65.0 mm and 113.0 mm height; the smallest radial clearance is 1.54 mm (BUZZER); pairwise interference check: 0 intersections. Module outlines marked PRELIMINARY must be re-checked against the actual parts once obtained.

6.5 Telemetry timing and regulatory figures

Table 6-4 Telemetry timing and regulatory figures

Quantity	GFSK baseline	LoRa option
Bytes / symbols on air	55 bytes	78 payload symbols + 12.25 preamble
Time on air (44-byte payload)	11.46 ms	92.4 ms (Semtech formula [S24])
Duty cycle at 1 packet/s	1.15 %	9.2 % (exceeds 2.5 %)
Occupied bandwidth	≈118 kHz (Carson)	125 kHz
ERP (+14 dBm, −2 dBi antenna)	≈10 mW	≈10 mW
Limits, GSR 853(E) Table II	500 mW ERP, ≤200 kHz, ≤2.5 %	same

6.6 Link budget

Free-space path loss: FSPL = 20·log10(d) + 20·log10(f) − 147.55 dB. Assumptions: +14 dBm, −2 dBi payload antenna, +1 dBi ground antenna, 1 dB losses.

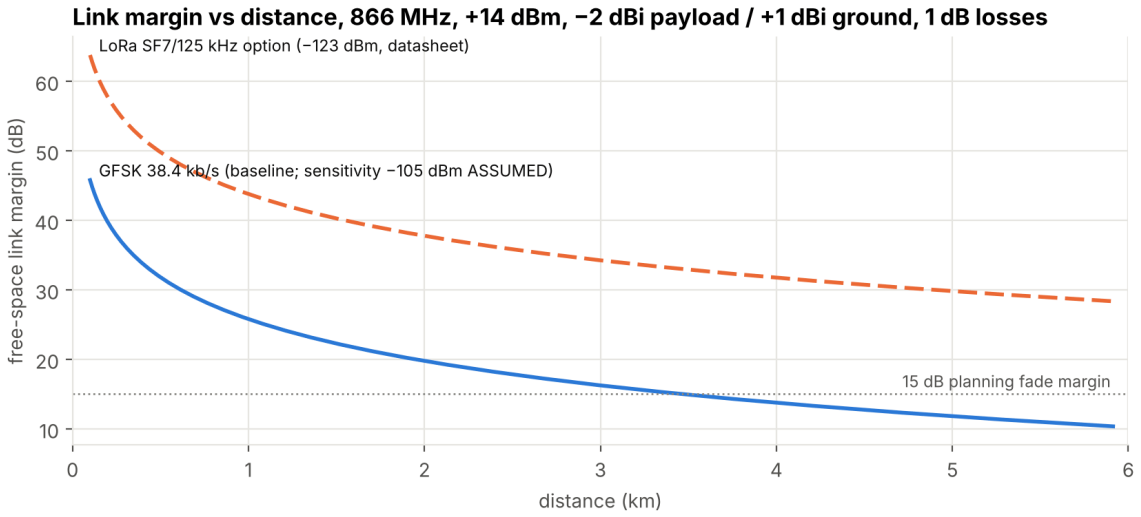


Figure 6-4 Free-space margin vs distance. GFSK sensitivity is ASSUMED (−105 dBm) pending a range test; LoRa sensitivity from the datasheet.

At 1 km the free-space received power is ≈-79 dBm, giving ≈26 dB margin for GFSK (assumed sensitivity) and ≈44 dB for LoRa SF7. A 15 dB fade margin is met to ≈3 km for GFSK in free space; tumbling, polarisation and ground effects will reduce this.

6.7 Recovery calculations

Steady descent: $v = \sqrt{(2mg / (\rho \cdot C_D \cdot A))}$. With m = 0.325 kg, v = 9 m/s, C_D = 0.75 (ASSUMED; flat circular canopies are commonly quoted at ≈0.75–0.80, e.g. Knacke's *Parachute Recovery Systems Design Manual* — not re-verified here) and ρ = 1.225 kg/m³: A = 0.0857 m², **D = 0.33 m**.

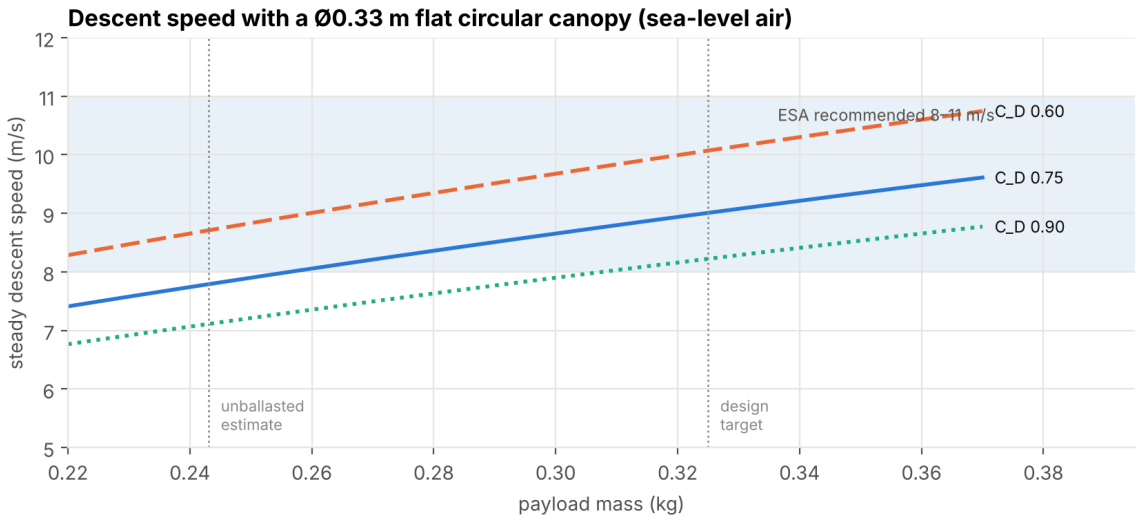


Figure 6-5 Descent speed vs mass for the chosen Ø0.33 m canopy across the assumed C_D range.

Attachment strength for a 500 N proof pull: each rod carries 250 N (A2-70 M3 yield ≈2.26 kN, factor ≈9); the yoke sees M = 250 N × 9 mm = 2,250 N·mm, Z = 10 × 4²/6 = 26.7 mm³, σ ≈ 84 MPa against ≈276 MPa yield for 6061-T6 (factor ≈3.3). A single central hole would raise the yoke stress about 2.8×, hence the two-leg bridle. Estimated opening load at an assumed 25 m/s deployment: ≈49 N. The PETG cap under the rod nuts must be proven by a pull test.

7 Bill of materials and cost

Prices were read from supplier pages on 24 Oct 2026 [S33] and are converted to INR including GST (18 % ASSUMED where pages show prices excluding GST). **Nothing has been ordered.** Full details, URLs, stock notes and alternatives are in STRATOS_BOM.xlsx.

Table 7-1 Compact BOM (key parts)

Item	Component	Model	Supplier	Cost incl. GST	Availability
P01	ESP32 development board	Espressif ESP32-DevKitC-32E (ESP32-WROOM-32E, 4 MB)	Mouser India	₹1,151	107 in stock
P02	Temperature / humidity / pressure sensor	7Semi BME280 nano breakout (Bosch BME280), 3.3 V I2C	Robocraze	₹500	In stock (57)
P03	6-DoF IMU	Adafruit 4438 LSM6DSOX 6-DoF breakout	Mouser India	₹1,347	121 in stock
P04	Sub-GHz radio module	Ai-Thinker Ra-01H (Semtech SX1276, 803–930 MHz)	Electronicscomp	₹648	86 available
P06	868 MHz antenna	Helix antenna 868 MHz, direct solder (≈1.5 dBi per listing)	Electronicscomp	₹58	1 available
P07	microSD breakout, 3 V	Adafruit 4682 Micro SD SPI/SDIO breakout – 3 V only	Mouser India	₹395	542 in stock
P08	microSD card	SanDisk 16 GB microSD Class 10	Electronicscomp	₹454	OUT OF STOCK at check
P09	3.3 V buck-boost regulator	Pololu S7V8F3 (#2122)	Pololu (import)	₹876	Active; backorders allowed
P10	18650 Li-ion cell	Molicel INR-18650-P28A 2800 mAh (authenticity to be verified on receipt)	Robu (mirror page)	₹529	In stock (mirror page; check live robu.in)
P11	Single 18650 holder	Keystone 1043 (PCB, THM)	Mouser India	₹337	16,493 in stock
P12	Cell protection module	TP4056 1 A USB-C charger with DW01A/FS8205A protection	Electronicscomp	₹19	460 available
S01	PETG filament 1.75 mm, 1 kg	Anycubic PETG	Robocraze	₹902	In stock (17)

Table 7-2 Cost summary

Category	INR incl. GST
Payload – electronics	₹4,608
Payload – power	₹1,786
Payload – passives	₹225
Payload – interconnect	₹524
Structure	₹902
Recovery	₹843
Ground station	₹1,940
Development equipment	₹243
Spares	₹2,131
Allowances (fasteners, Al bar, shipping, Pololu import)	₹3,450
Subtotal	₹16,652
Contingency 15 %	₹2,498
Total estimated procurement	₹19,150
Per-payload production estimate (flight unit, incl. import allowance)	₹11,053
Ground station (separate; laptop assumed owned)	₹1,940

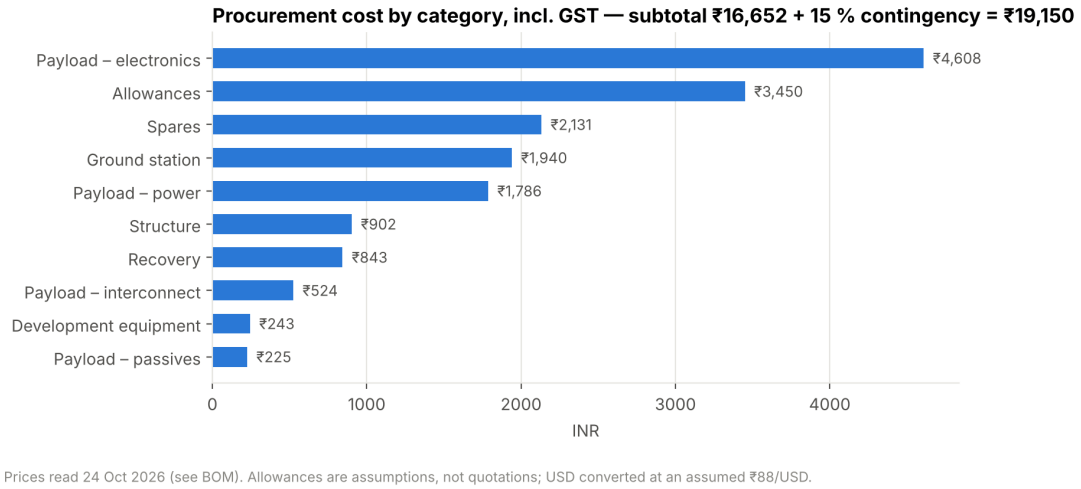


Figure 7-1 Procurement cost by category.

Procurement notes

- **Pololu S7V8F3** is not stocked by the Indian retailers checked; it is priced at USD 9.95 with an assumed ₹88/USD and a ₹2,500 shipping/duty allowance. A domestic LDO alternative is listed but not priced.
- **18650 cell:** the Robu 'Samsung 35E' listing states its cells are copies — do not use. A Molicel P28A is priced as a candidate; authenticity and capacity must be verified on receipt.
- **microSD:** the 16 GB card was out of stock on 24 Oct; industrial cards were unavailable or expensive. Test every card before use (risk R04).
- **Ra-01H** has stamp pads and no IPEX connector: the payload helix is soldered to the adapter; the ground pigtail's coax must be soldered to the ANT pad.
- Robu 'mirror' pages showed lower prices than live robu.in pages; Mouser prices exclude GST and possibly duties. Re-check on the purchase day.
- Items already owned (laptop, multimeter, 3D printer, soldering station) are ASSUMED and listed as 'Owned (to confirm)'.

8 Development and verification

8.1 Development roadmap

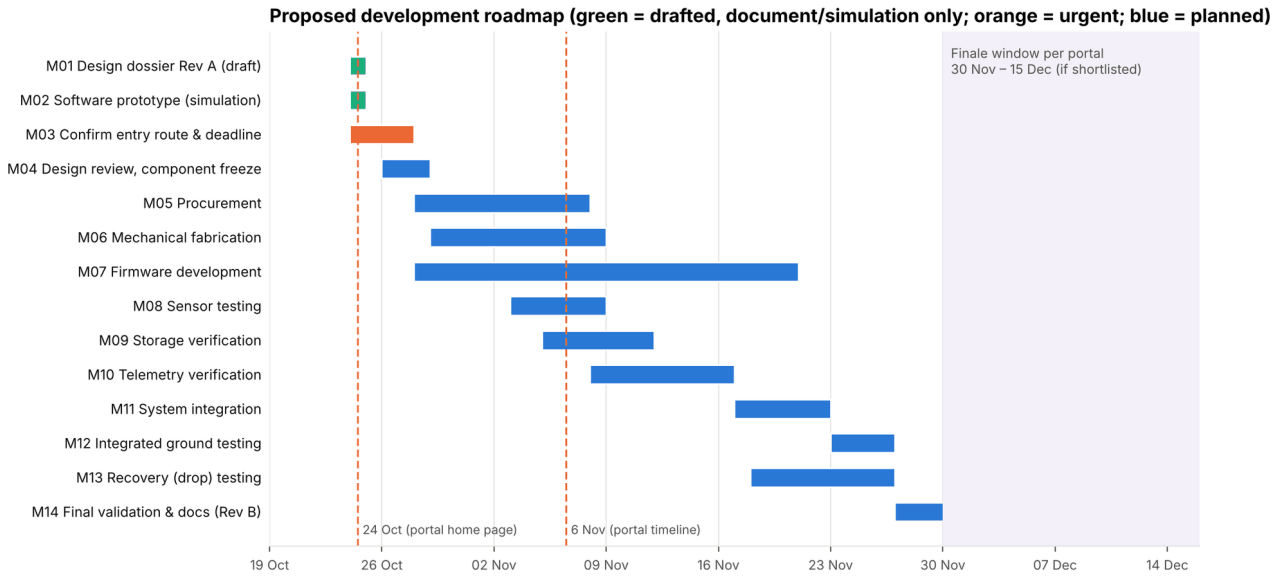


Figure 8-1 Proposed roadmap from 24 Oct 2026. Only M01 (this dossier, a draft pending team review) and M02 (software prototype, simulation only) exist; everything else is planned.

Schedule conflict and recovery plan. If the submission window closes on 24 October, the concept must be submitted now with this dossier as evidence of design maturity. If shortlisted, the realistic target before the finale (30 Nov–15 Dec) is a **working bench prototype** that logs, transmits and reconciles real sensor data; no flight is planned before the finale. If the team is not eligible this cycle, the roadmap continues unchanged toward the next edition or a CanSat competition with published rules.

8.2 Test and verification matrix

Table 8-1 Verification of every requirement (acceptance criteria fixed before testing)

ID	Method	Acceptance criterion	Status
FR-01	Test	Valid readings every record for 30 min; offsets vs reference recorded (no pass/fail on accuracy until calibration plan agreed)	Not started
FR-02	Test	|a| = 1.00 ± 0.05 g at rest in each orientation; gyro sign and axis correct	Not started
FR-03	Test	rec_seq continuous; 95 % of intervals within 100 ± 5 ms; overruns flagged	Not started
FR-04	Test	All records present on microSD; flush interval ≤ 1 s	Not started
FR-05	Test	Every damaged line reported (ONBOARD_LINE_INVALID); no damaged line accepted	Demonstrated in software simulation (truncated line, synthetic) — hardware test not started
FR-06	Test	≥ 99 % of packets received; every packet CRC-valid; pkt_seq continuous	Not started
FR-07	Test	Every corrupted frame rejected and logged with raw bytes; every accepted row has rx_utc	Demonstrated in software simulation (synthetic data) — hardware test not started
FR-08	Test	Every skipped pkt_seq listed in ground_gaps.csv; reboot flagged SESSION_CHANGE	Demonstrated in software simulation (synthetic data) — hardware test not started
FR-09	Test	All induced losses classified correctly; zero unexplained discrepancies	Demonstrated in software simulation (synthetic data) — hardware test not started
FR-10	Test	INVALID flag set and sentinel/empty field written; zero values never used for absent data	Demonstrated in software simulation (synthetic data) — hardware test not started
FR-11	Analysis + Test	Measured average current × 4 h ≤ 50 % of measured usable capacity	Analysis complete (this dossier, Rev A) — hardware test not started
FR-12	Analysis + Inspection	All rigid parts inside Ø 66 × 115 mm; mass within the published limit	Analysis complete (this dossier, Rev A) — hardware test not started
FR-13	Analysis + Test	Mean of 3 drops within the target band	Analysis complete (this dossier, Rev A) — hardware test not started
FR-14	Demonstration	Beeper audible at 30 m outdoors	Not started

ID	Method	Acceptance criterion	Status
FR-15	Inspection + Analysis	Frequency, bandwidth and power inside the notified limits	Analysis complete (this dossier, Rev A) — hardware test not started
FR-16	Test + Inspection	No part detached or loosened; no connector unseated	Not started
FR-17	Test + Inspection	Protection opens; PTC trips; firmware enters safe shutdown at 3.30 V	Not started
FR-18	Test	Log file closed; final line CRC-valid; SAFE SHUTDOWN logged	Not started

8.3 Planned test procedures (summary)

Table 8-2 Planned test stages

Stage	Test	Evidence
1 Design review	Rules, envelope, interfaces, BOM	Signed checklist
2 Component/interface	Chip IDs (BME280 0x60), cell capacity, regulator output, switch rating	Incoming-inspection sheet
3 Sensors	30 min logging next to reference instruments; six-orientation IMU test	Serial logs, comparison sheet
4 Local logging	4 h run; 10 random power cuts	Log files; reconciliation of onboard lines
5 Telemetry	Bench link then range test at 100 m–1 km; channel survey first	Transmitter + receiver logs; packet-error rate vs distance
6 Data integrity	Deliberately skipped sequence numbers and corrupted frames from a test transmitter	Reject/gap logs; unit tests
7 Integrated ground test	4 h run with induced faults, measured current	Reconciliation report; INA219 log
8 Mechanical/recovery	Shake, drop from a fixed structure (3 drops timed), 500 N pull	Inspection record, video, timing
9 Mission validation	Only with written permissions and a safety review	Authorised plan, post-test report

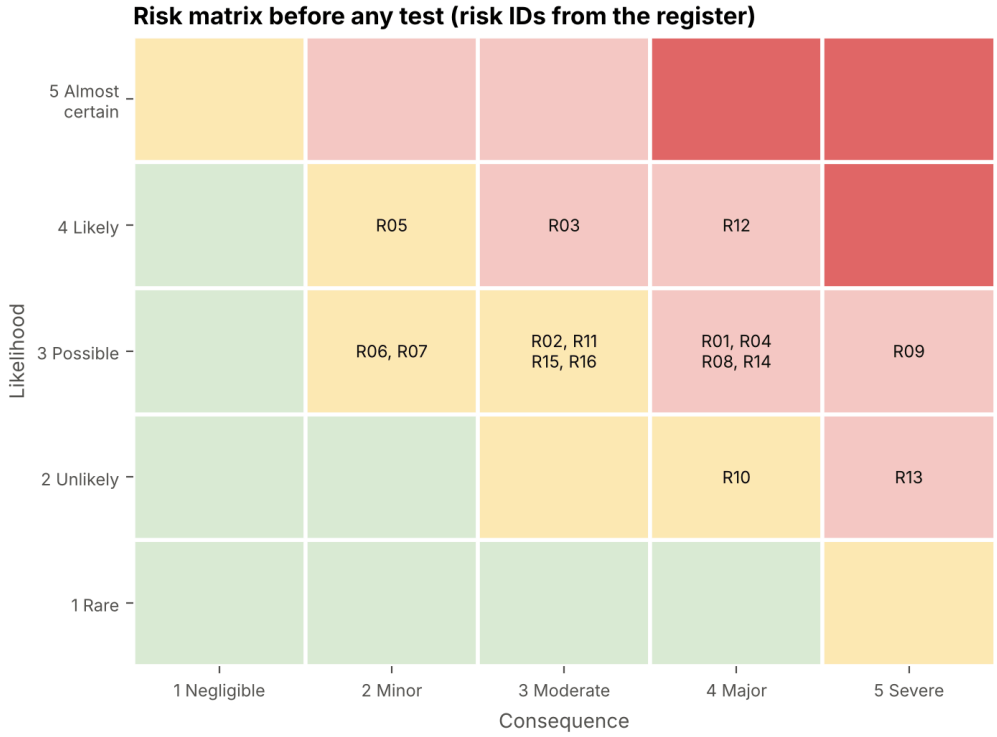
8.4 Software evidence available now

- `python -m unittest discover -s tests` — 8 tests pass: CRC check value; packet round-trip; **every single-bit error in a frame detected**; wrong length rejected; absent ≠ zero; log-line round-trip and truncation; gap/corruption/duplicate/session tracking; simulated reconciliation classifies all injected faults.
- The C packet/log-line code compiles with `-Wall -Wextra -Werror` and its output is byte-identical to the Python reference for four test records, including sentinels and extreme values.
- These are software results on synthetic data; they do not verify any hardware requirement.

8.5 Known limitations

- No sensor accuracy, timing jitter, power, radio range, SD reliability or structural result is measured yet.
- IMU accelerometer range (±16 g) saturates above 16 g; launch shocks above that will clip (survival is a separate matter).
- No real-time clock: absolute time exists only at the ground; onboard time restarts after a reset (made visible by the session ID).
- Payload antenna performance inside the can is unknown; the link budget uses assumed gains and an assumed GFSK sensitivity.
- Adaptive power control (GSR 853(E) Table II) is not implemented on the one-way link.

9 Risk and safety



Bands: Low 1–4, Medium 5–9, High 10–16, Critical 17–25. No mitigation has yet been verified by a hardware test.

Figure 9-1 Risk matrix before any test. No mitigation has yet been verified by a hardware test.

Table 9-1 Risk register (condensed; full register with causes and owners in STRATOS_Risk_Register.xlsx)

ID	Risk	L	C	Score	Mitigation (proposed)	Verification	Status
R01	Power interruption	3	4	12	Locking JST-XH connectors; cell retained by holder + zip tie; DPDT switch recessed in bottom cap; 100 µF bulk capacitor; 1 s flush limits loss; line CRC detects truncation; new session ID makes the reset visible	Bench power-cut test (10 cuts at random times): log recoverable, loss ≤ 10 records per cut, every truncated line flagged	Proposed
R02	Sensor malfunction	3	3	9	Boot self-test with chip-ID check (BME280 ID 0x60 distinguishes it from a BMP280); per-field INVALID flags and sentinels; re-initialisation attempt every 10 s; spare modules	Disconnect each sensor during a bench run: flags set, logging continues, sensor recovers on reconnection	Proposed
R03	Invalid (biased) measurements	4	3	12	BME280 on the opposite face of the spine from the ESP32, beside shell vent slots; low BME280 duty; plausibility limits; comparison against reference instruments before use	Side-by-side comparison with a reference thermometer/hygrometer and barometer, inside the closed shell, for 30 min; record offset and settling time	Proposed
R04	Storage failure	3	4	12	Test every card on receipt (full-capacity write/read check) and reject counterfeits; industrial-grade card preferred when one can be sourced (none in stock on 24 Oct 2026, so the BOM lists a consumer card); card secured with polyimide tape; one file per session; 1 s flush; line CRC; telemetry provides an independent second copy of 1 record in 10	4 h soak test, power-cut test and drop test with the card installed; card readable and all lines CRC-valid	Proposed
R05	Packet loss	4	2	8	Large link margin (see link budget); LoRa SF7 at 1 packet per 4 s as a longer-range fall-back that stays within the 2.5 % duty limit; elevated ground antenna; sequence numbers make every gap visible; onboard log recovers missing records	Range test at increasing distances: log packet-error rate, RSSI and SNR; reconciliation explains every gap	Partly shown in simulation
R06	Corrupted packets	3	2	6	Radio packet-engine CRC plus application CRC-16 checked before any field is trusted; rejected frames logged with raw bytes	Software: every single-bit error in a frame rejected (unit test, synthetic). Hardware: inject corrupted frames from a test transmitter	Partly shown in simulation
R07	Radio interference	3	2	6	Listen on the channel before each test and choose the quietest; private sync word; magic bytes, length and CRC reject foreign frames	Channel survey log before every radio test	Proposed
R08	Mechanical damage	3	4	12	Load path through two M3 stainless rods; ballast and battery low in the can; threadlocker / nyloc nuts; zip-tie retention; descent rate limited by parachute sizing	Drop and shake tests at increasing severity; inspection checklist after each test	Proposed
R09	Recovery-system failure	3	5	15	Parachute sized by calculation with margin; swivel to prevent line twist; bridle and yoke rated far above	Repeated drop tests from a controlled height measuring descent rate; pull test of yoke and	Proposed

ID	Risk	L	C	Score	Mitigation (proposed)	Verification	Status
					opening load; packing procedure; no flight until drop tests pass and permission is held	bridle to ≥ 10× estimated opening load	
R10	Loss of the payload	2	4	8	Recovery beeper after landing detection; last RSSI and descent data from telemetry; high-visibility parachute; contact label; wind limit for tests; optional GNSS (deferred, pins reserved)	Beeper audibility test at 30 m; recovery drill	Proposed
R11	Component procurement delay	3	3	9	Order all long-lead items in the first week; two suppliers identified per critical part; spares in BOM; listed alternatives	Procurement tracker reviewed weekly	Proposed
R12	Insufficient time for integration and validation	4	4	16	Staged plan with exit criteria; software pipeline built first (done in simulation); weekly milestone review; agreed de-scope order (GNSS, landing beeper, IMU) if behind	Milestone review against roadmap each week	Proposed
R13	Li-ion battery hazard	2	5	10	Genuine cell from reputable seller; DW01A/FS8205A protection; 2.5 A-hold resettable PTC fuse; insulated holder; cell charged outside the can (external charger) or, supervised, through the module USB-C with the shell removed; firmware safe shutdown at 3.30 V	Inspection before each use; short-circuit test of protection on the bench with current limit	Proposed
R14	Regulatory non-compliance	3	4	12	Radio configured inside 865–867 MHz within notified limits (GFSK, 1.15 % duty, ≈10 mW ERP); lowest power that gives an adequate margin in the range test; confirm ETA / prototype status with the institution before any outdoor transmission; ground tests only until written permissions exist; no drone drops without compliance with Drone Rules	Configuration review and sign-off before each radio or flight test	Proposed
R15	Competition rules differ from assumed envelope	3	3	9	Parametric CAD driven by one baseline file; obtain official rules before fabrication	Rules compliance checklist completed before fabrication	Proposed
R16	Counterfeit or mislabelled parts	3	3	9	Reputable suppliers; chip-ID checks in self-test; capacity test of each cell before use	Incoming inspection: chip IDs read, cell capacity measured	Proposed

Scales: likelihood 1 Rare (< 5 %) … 5 Almost certain (> 80 %); consequence 1 Negligible … 5 Severe (injury, payload loss or project failure). Bands: Low 1–4, Medium 5–9, High 10–16, Critical 17–25.

9.1 Recovery and deployment considerations

- No deployment from aircraft (drones, balloons, kites) without written permission; Rule 26 of the Aircraft Rules prohibits dropping articles and parachute descents.
- Drop tests from a fixed structure over a cordoned area, with a spotter, low wind and no people below.
- Inspect bridle, swivel, yoke and nuts before every test; replace any line showing abrasion.
- High-visibility canopy, contact label and the landing beeper aid recovery.

9.2 Radio permissions

- Operate only in 865–867 MHz within GSR 853(E) limits; channel survey before every test; lowest adequate power.
- Confirm with the institution whether the prototype needs ETA or falls under an R&D/prototype exemption before outdoor use (risk R14).

9.3 Battery safety

- Genuine cell, DW01A protection, PTC fuse, insulated holder, firmware cut-off at 3.30 V.
- Charge outside the can in a charger on a non-flammable surface, never unattended; supervised USB-C bench charging only with the shell removed.
- Store at partial charge; discard any swollen, dented or hot cell.

Appendix - Drawing and schematic sheets

Reduced copies; the full-resolution sheets are in STRATOS_Mechanical_Drawings.pdf and STRATOS_Electronics_Schematic.pdf.

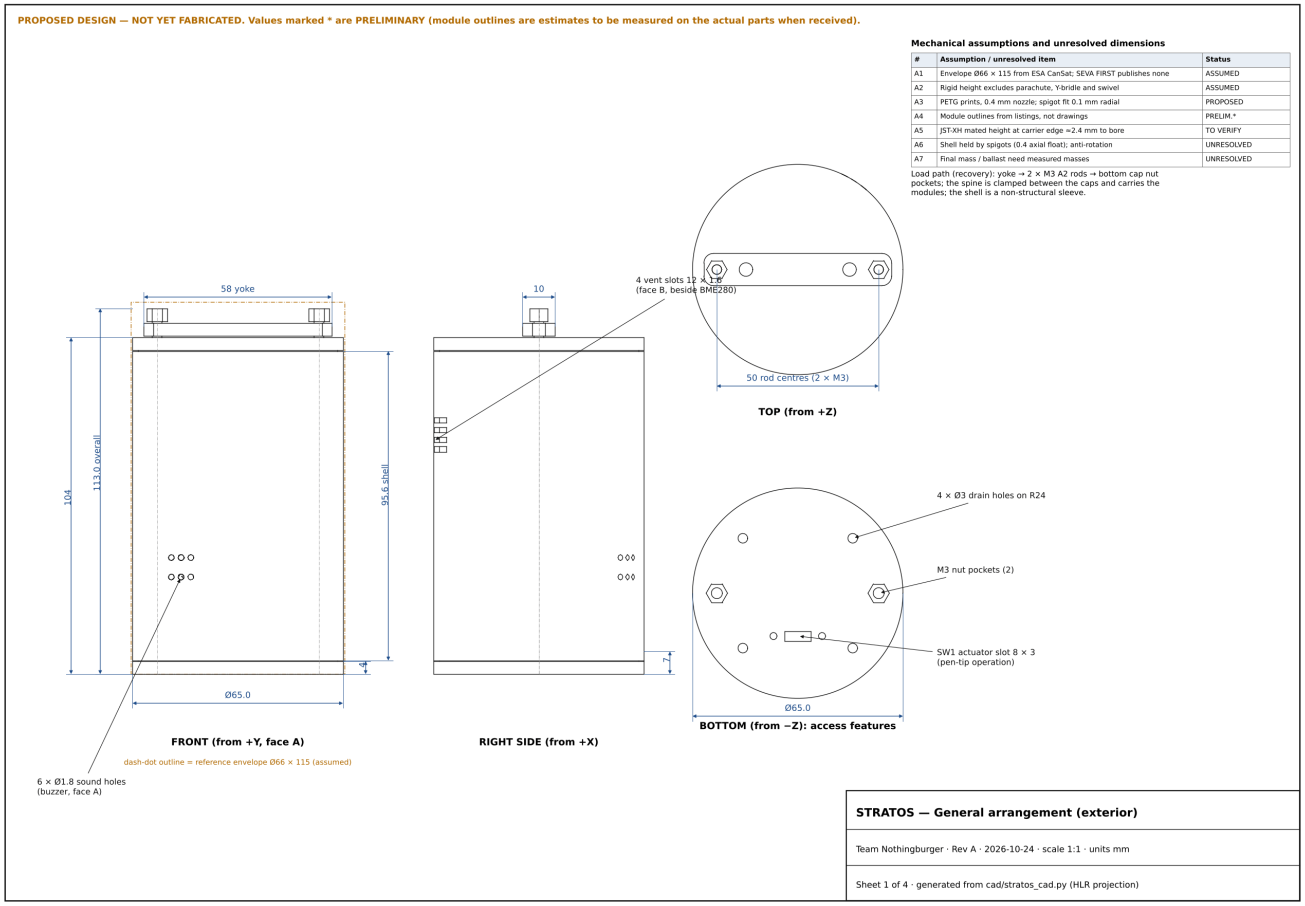


Figure F-1 Mechanical drawing sheet 1: General arrangement.

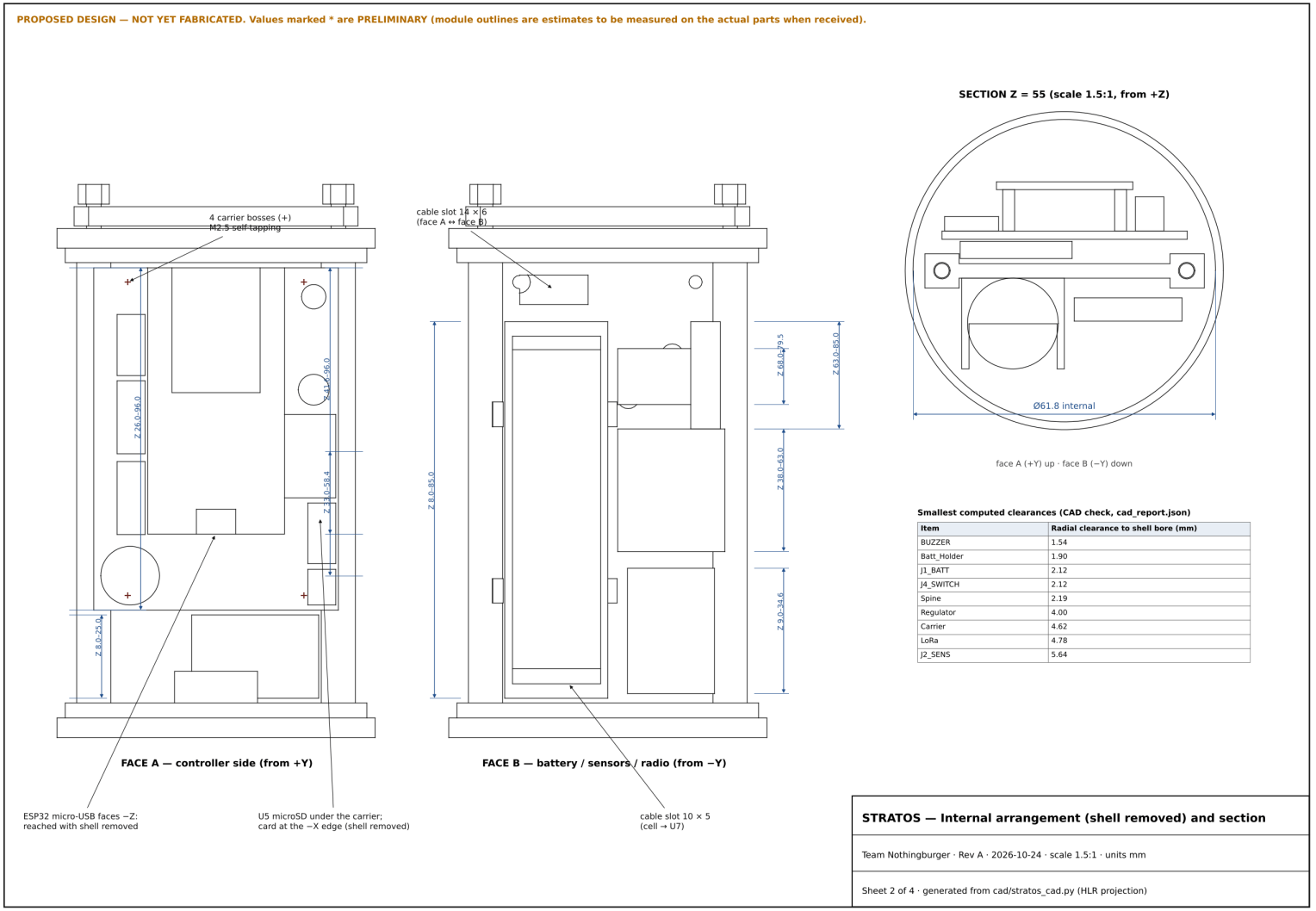


Figure F-2 Mechanical drawing sheet 2: Internal arrangement and section.

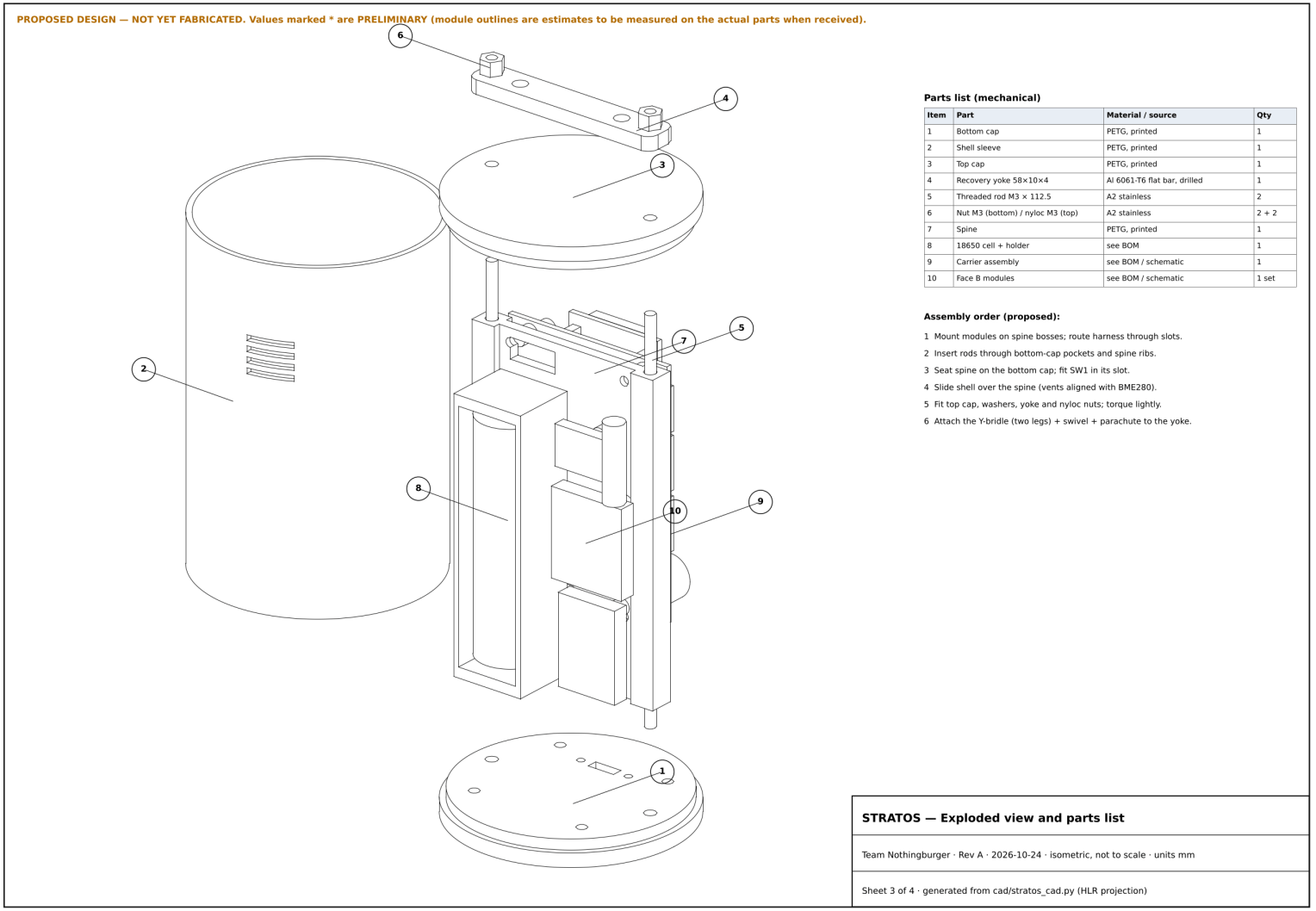


Figure F-3 Mechanical drawing sheet 3: Exploded view and parts list.

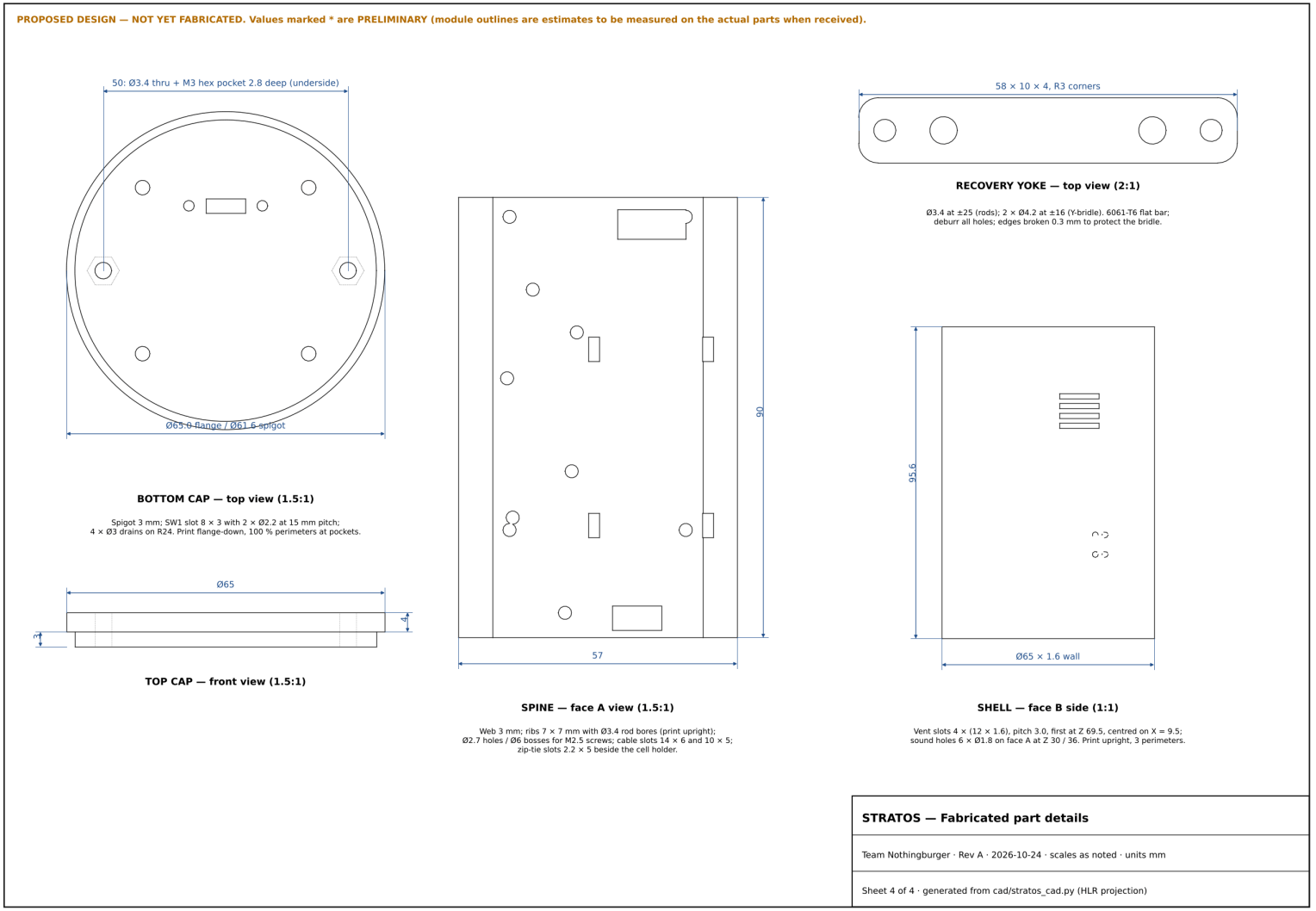


Figure F-4 Mechanical drawing sheet 4: Part details.

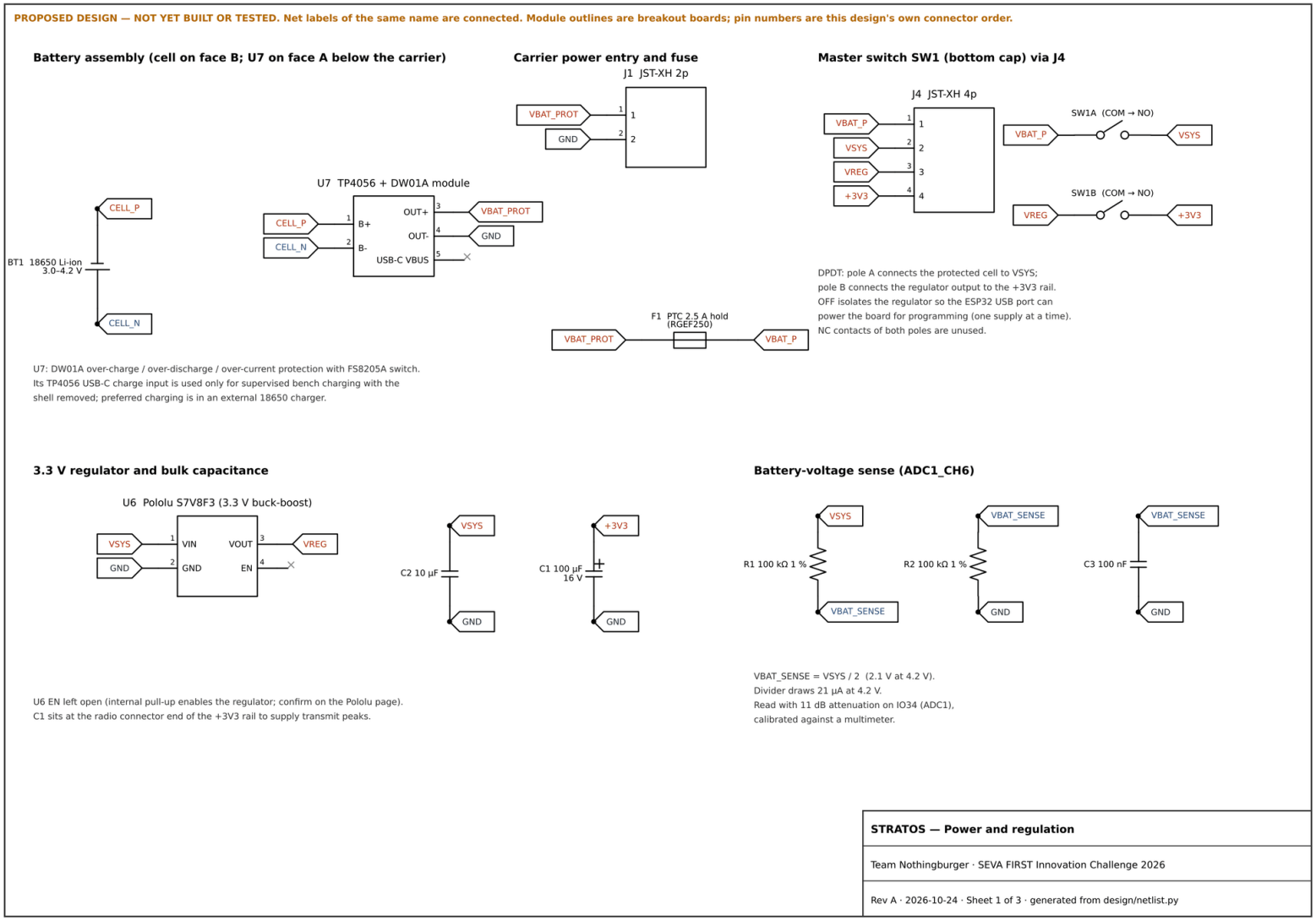


Figure F-5 Schematic sheet 1: Power and regulation.

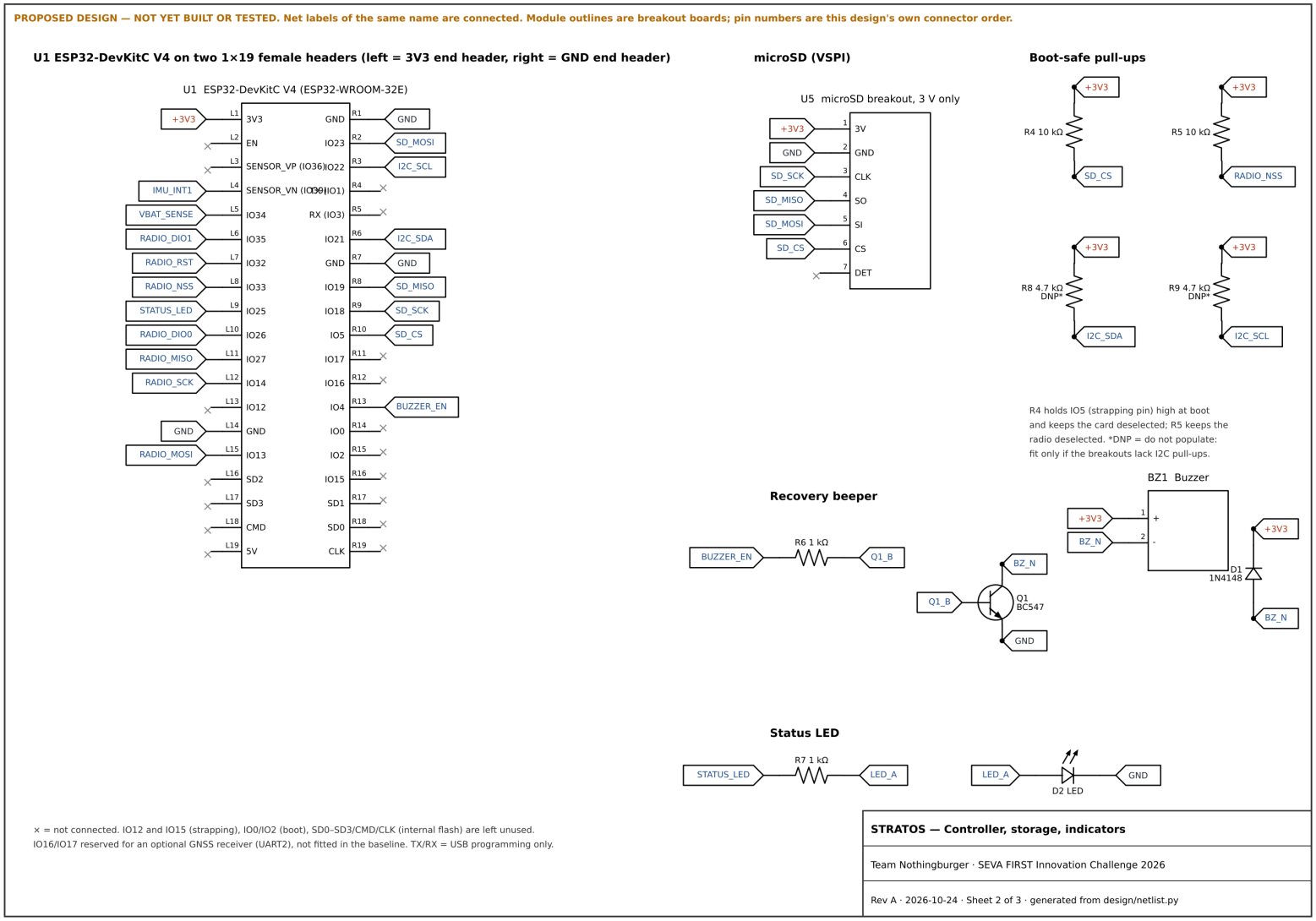


Figure F-6 Schematic sheet 2: Controller, storage, indicators.

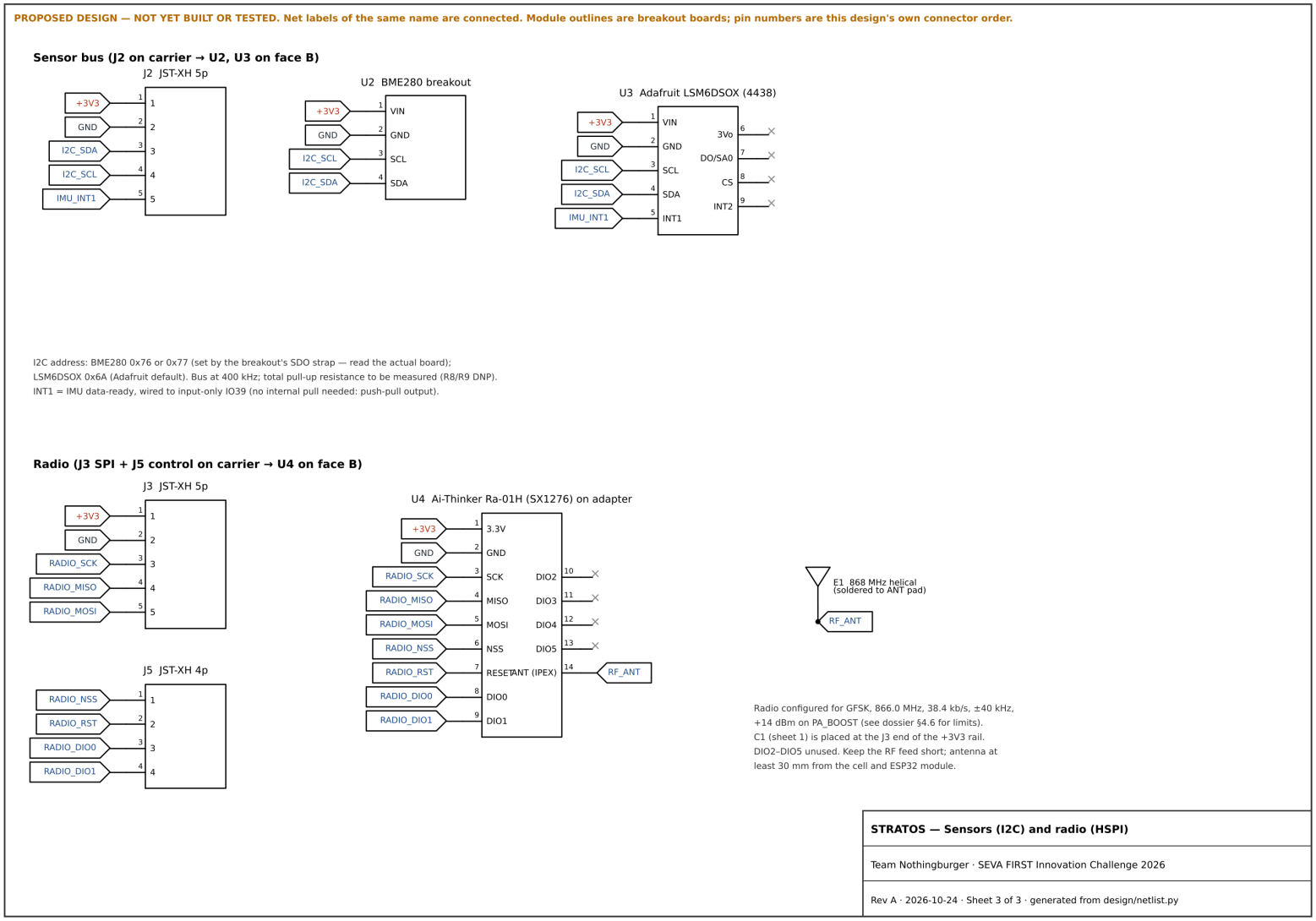


Figure F-7 Schematic sheet 3: Sensors and radio.